

No Key? No PIN? No Combo? No Problem!

P0wning ATMs For Fun and Profit

Roy Davis



roydavis@roydavis.org



[@hack_all_things](https://twitter.com/hack_all_things)



[/in/roy-davis](https://www.linkedin.com/in/roy-davis)





Why ATMs?



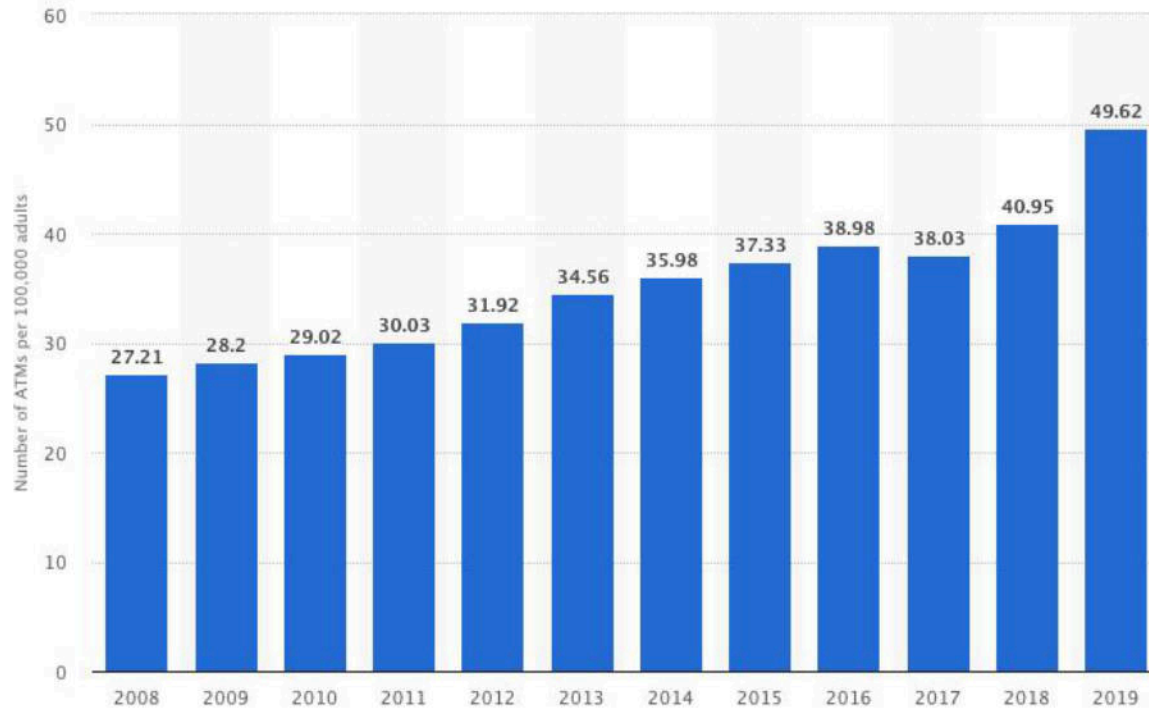


<https://www.youtube.com/watch?v=ZO956U10Zlc>



Number of ATMs globally from 2008 to 2019

(per 100,000 adults)



© Statista 2021

<https://www.statista.com/statistics/741708/number-of-atms-worldwide/>



*mini*ATM and Scrip Communications Protocol And Message Format Specification



PRELIMINARY

VERSION TSCD2.9

February 4, 2002

Triton Systems of Delaware, Inc.
522 East Railroad
Long Beach, MS 39560
PHONE: (228) 868-1317
FAX: (228) 868-0437

Copyright Notice

<https://bit.ly/3ySNvey>





Barnaby Jack

DEF CON 18 - 2010

Westin Hecker

DEF CON 24 - 2016

Alexey Osipov
Olga Kochetova

Blackhat Europe - 2014

Mike Davis

DEF CON 27 - 2019

Trey Keown
Brenda So

DEF CON 28 - 2020

Alexander Forbes

Disobey - 2019

Plore

DEF CON 24 - 2016



On tap...

ATM Acquisition

Damage Inc.

Some general ATM info

Licensed to operate (and sniff traffic??)

Picking the ATM case lock

Resetting the admin password

Bypassing the Electronic Vault Lock

Q&A



The Goal...

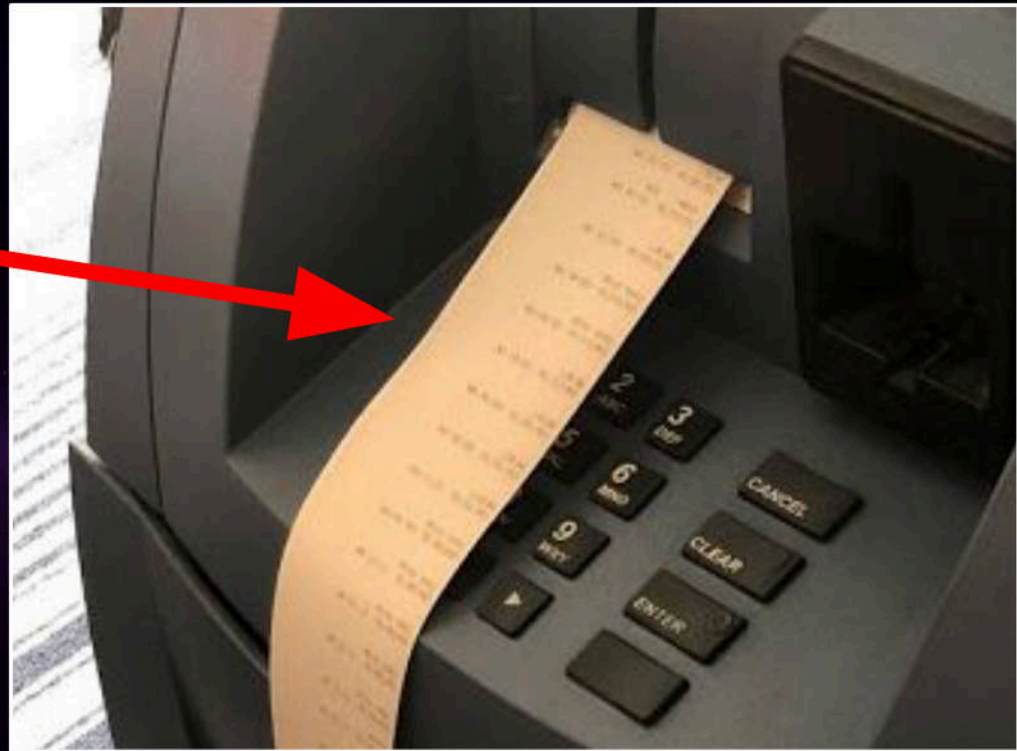


Atm machines - \$100 (Hastings)

image 5 of 5







Alerts

Monitor the web for interesting new content

🔍 Create an alert about...

My alerts (11)



ATM Auction Minnesota



Dickey's Barbeque, Gas Station, and Car Wash - Cambridge, MN

10% BUYERS FEE - OPEN HOUSE: Friday, January 8th (11:00am - 2:00pm); PICK UP: Tuesday, January 12th (10:00am - 2:00pm), Wednesday, January 13th (10:00am - 2:00pm) Sign up for an appointment:

<https://www.signupgenius.com/go/PikeAuctionDickeys>

🕒 Dec 29 @ 2:43pm CST (Start)

🕒 Jan 11 @ 6:00pm CST (End)

📅 Add to Calendar

📍 4000 Main Street South, Cambridge, MN 55008, US (map)

[View Catalog and Bid!](#)

[View All Photos](#)
[Details](#)
[Map](#)
[Terms](#)
[All Photos](#)

Online Only Auction Ends, January 11th, 2021 at 6:00 p.m.



Unknown Working Condition

#1709: Hyosung ATM Machine



Unknown Working Condition

CONFIRM BID
\$1

CANCEL

☆ Ends 12d 5h 36m

Bids: TBD

● Asking [\$1]

Your max [not set]

CONFIRM BID
\$1



#1709: Hyosung ATM Machine - No Code for Cash Box



SOLD

Bids: 33

YOU WON

● Your bid [\$220]



Nautilus Hyosung 2700 CE

\$2,460.00

[ATM Mega Store](#)

Delivery by Thu, Jul 15

[Home](#) / [Shop](#) / [New ATMs](#) / [Nautilus Hyosung NH 2700CE ATM](#)

Sale!



Nautilus Hyosung NH
2700CE ATM

~~\$2,480.00~~ **\$2,299.00**



HOME

GROCERIES

FUEL

CAR WASH

CAMBRIDGE BP

BAIT

DICKEY'S

CONTACT

DICKEY'S BARBECUE PIT GRAND OPENING FEBRUARY 1ST, 2018!



Dickey's Franchisee Brings Texas-Style Barbecue to Cambridge, MN

March 15, 2018 · ·

Share 0

Tweet

Share

Share

1



Dickey's Franchisee Brings Texas-Style Barbecue to Cambridge, MN

March 15, 2018 · ·

Cambridge, MN (RestaurantNews.com) New Dickey's Barbecue Pit franchisee, [REDACTED] plans to open his first Dickey's Barbecue Pit location to Cambridge, MN in Spring 2018.

"At Dickey's Barbecue Pit, we pride ourselves on serving each and every guest authentic, Texas-style barbecue," says Laura Rea Dickey, CEO of Dickey's Barbecue Restaurants, Inc. "With new franchisees such as Scott that truly love the brand, we can continue our growth and more importantly continue serving our guests delicious barbecue."

DICKEY'S
BARBECUE PIT



BUSINESS

Recent bankruptcy filings in Minneapolis and St. Paul

MARCH 1, 2019 — 11:57PM

Minneapolis

[REDACTED], as surety for Main & 65 LLC, Cambridge Dickie's LLC and Cambridge Fuel Inc., doing business as Cambridge BP, 1015 Essex St. SE., Minneapolis; filed Feb. 27, 19-40531; Chap. 7; assets, \$43,545; liabilities, \$1,522,828.

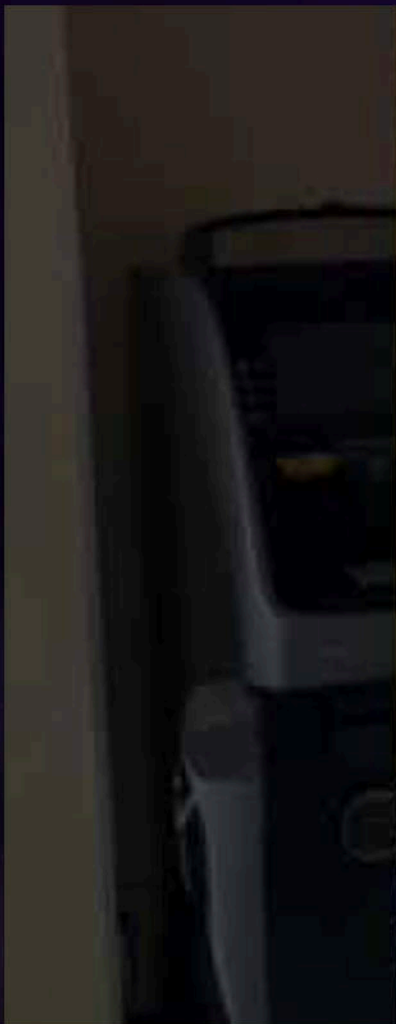




Photo by Sarv Mithaqiyani























Booting...

Microsoft Windows CE 6.0 [EBOOT Version V06.22.16]

(c)Copyright 2010 Nautilus Hyosung Inc. All Rights Reserved.



Windows Embedded CE 6.0



Found 8 open ports
(8083 scanned).

- **80** World Wide Web HTTP
- **443** http protocol over TLS/SSL
- 5555** Personal Agent
- 8001** VCOM Tunnel
- 8003** Mulberry Connect Reporting Service
- 8004**
- 8006** World Programming analytics
- 8010**

Source: Trey Keown & Brenda So - DEF CON 28

Target: 192.168.10.104

Command: nmap -p 1-65535 -T4 -A -v 192.168.10.104

Hosts

Services

OS

Host



192.168.10.104

Nmap Output

Ports / Hosts

Topology

Hosts

nmap -p 1-65535 -T4 -A -v 192.168.10.104

Completed NSE at 02:48, 0.00s elapsed
Nmap scan report for **192.168.10.104**
Host is up (0.00070s latency).

Not shown: 65534 closed ports

PORT	STATE	SERVICE	VERSION
5555/tcp	open	freeciv?	

1 service unrecognized despite re

<https://nmap.org/cgi-bin/submit.c>

SF-Port5555-TCP:V=7.80%I=7%D=7/10

SF:LSessionReq,1,"\x04")%r(TLSSes

MAC Address: 00:A0:7C:41:A3:64 (T

Device type: media device

Running: Microsoft Windows Pocket

OS CPE: cpe:/o:microsoft:windows_

OS details: AT&T U-Verse set-top

Network Distance: 1 hop





OUT OF SERVICE

ATA4 is temporarily out of service.

ERROR CODE : 07005(04)

EPP ERROR



Thanks for another purchase, Roy! Your order is confirmed.

We'll let you know when it's on the way. In the meantime, we have more things you'll love at unbeatable prices!

[View order details](#)[Browse deals](#)

Order summary



Refurbished Nautilus Hyosung ATM Machine Black
Keypad 6000k + 6 Mo. Warr...

Total: \$322.11

Order number: [REDACTED]

Item ID: 254176010696



How do ATMs work?













04/2017 04:14:00

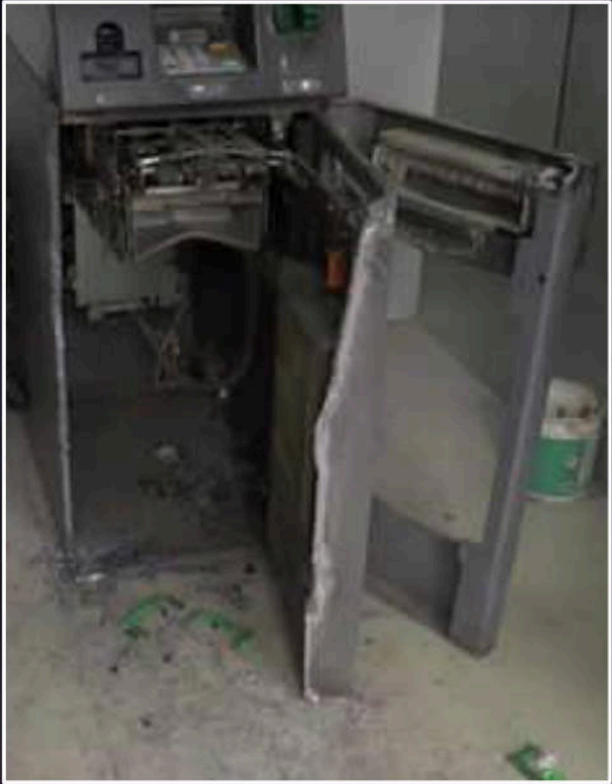




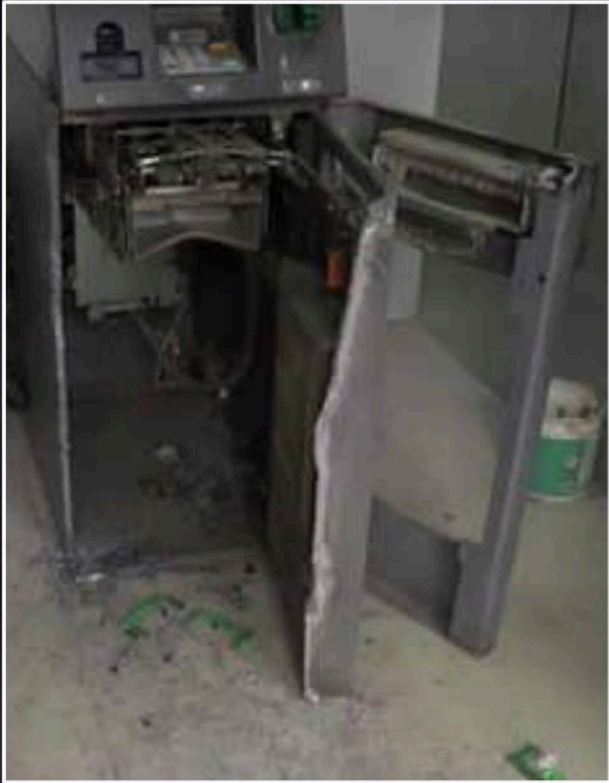




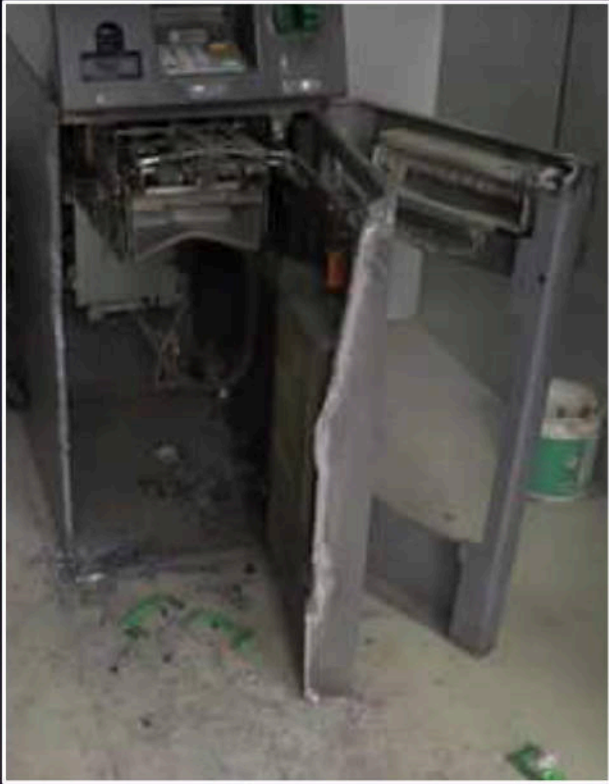
Damage, Inc.



Damage, Inc.



Damage, Inc.





NEW AT 6

THOUSANDS STOLEN FROM FOUR ATMS ACROSS THE METRO



6:00 44°





The ATM was damaged, but
its cash box remained intact



Camera (optional)

Touchscreen

Receipt
Dispense

Cash
Dispense



CPU Housing

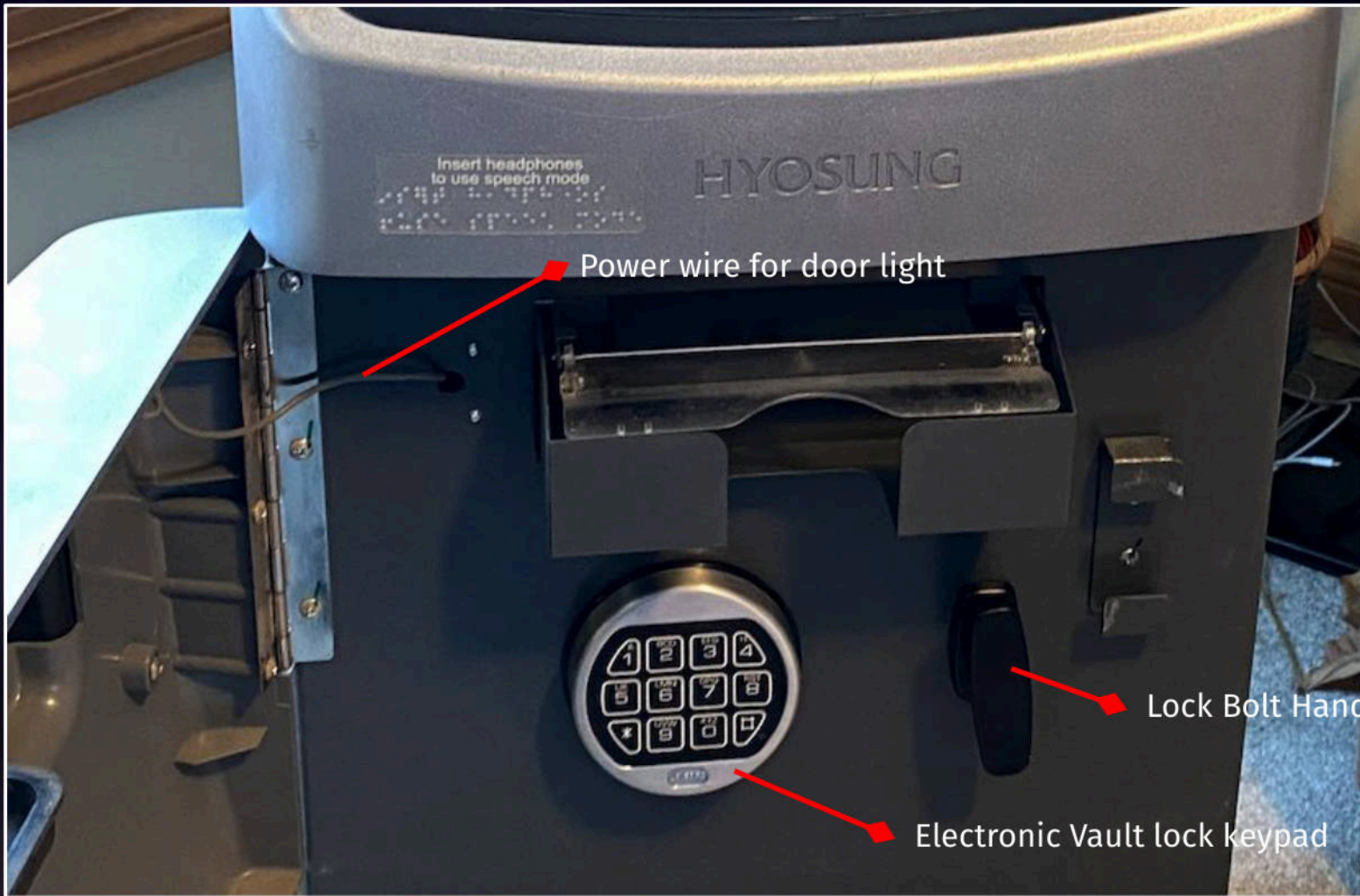
Card Reader (mag / chip)

Encrypting Pin Pad

Safe Door Cover

Vault Housing





Power wire for door light

Lock Bolt Handle

Electronic Vault lock keypad





High Security Electronic Lock, Type 1
Type 1
MODEL 4200M
SN 42001823100112K
UL LISTED
CLASS 1, 110V AC, 50/60 Hz
EN 1300 A, 50/60 Hz, 110V AC
EN 1300 B, 50/60 Hz, 110V AC
INSTRUCTION: <http://www.sds-mea.com>, T17-089

CLASS 2
M 110303
EN 1300
CLASS B

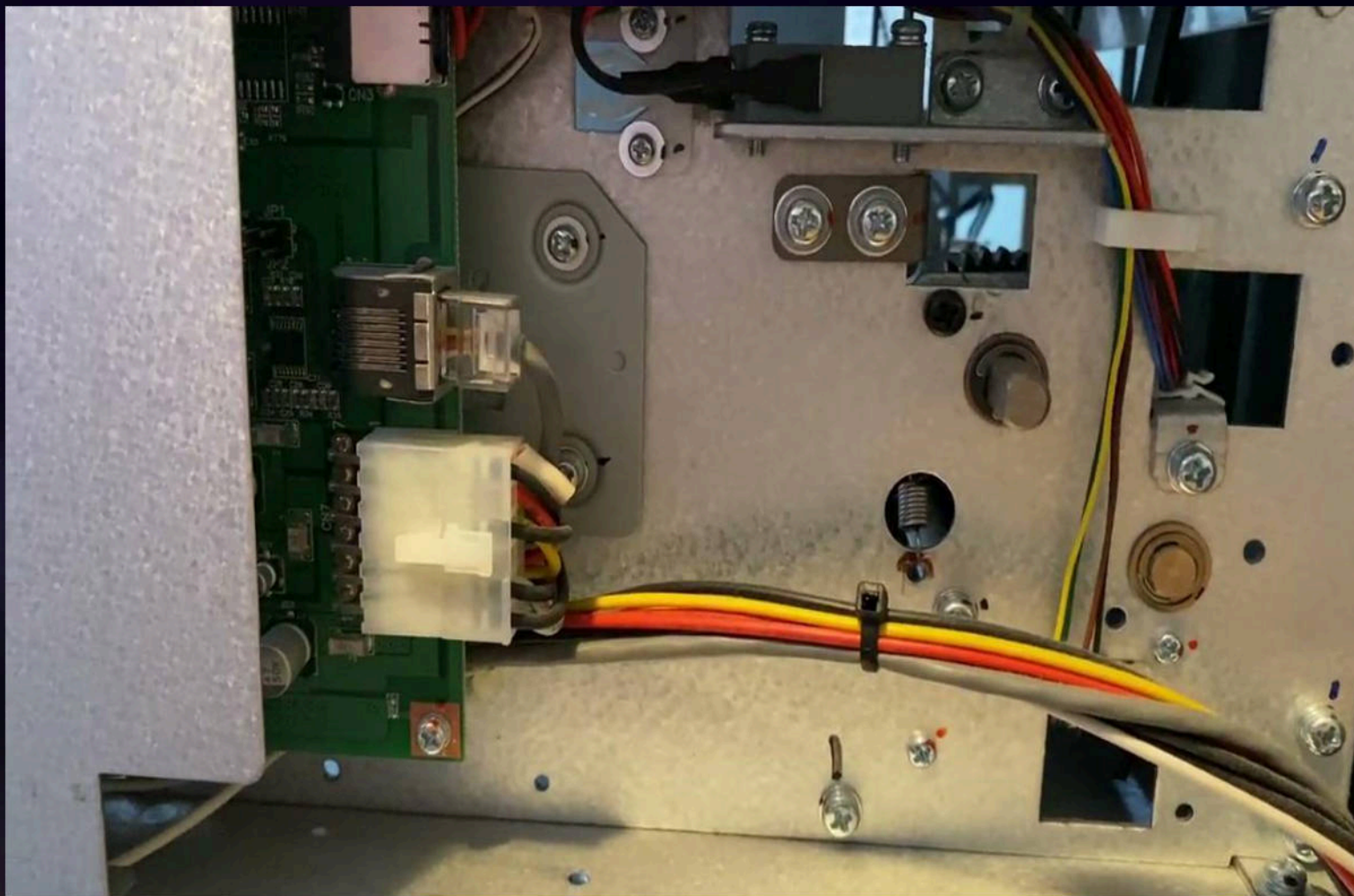
UL LISTED
CLASS 2
EN 1300
CLASS B

ID 1014

UL LISTED
CLASS 2
EN 1300
CLASS B











Cash Dispenser with 3 cassettes (Option)







Nautilus Hyosung ATM Machine New Bezel & Cassette Keys 1500 1800 2700 Halo Force

Condition: **New**

Quantity:

1

More than 10 available
64 sold / [See feedback](#)

Price: **US \$18.99**

Buy It Now

Add to cart

Best Offer:

Make Offer

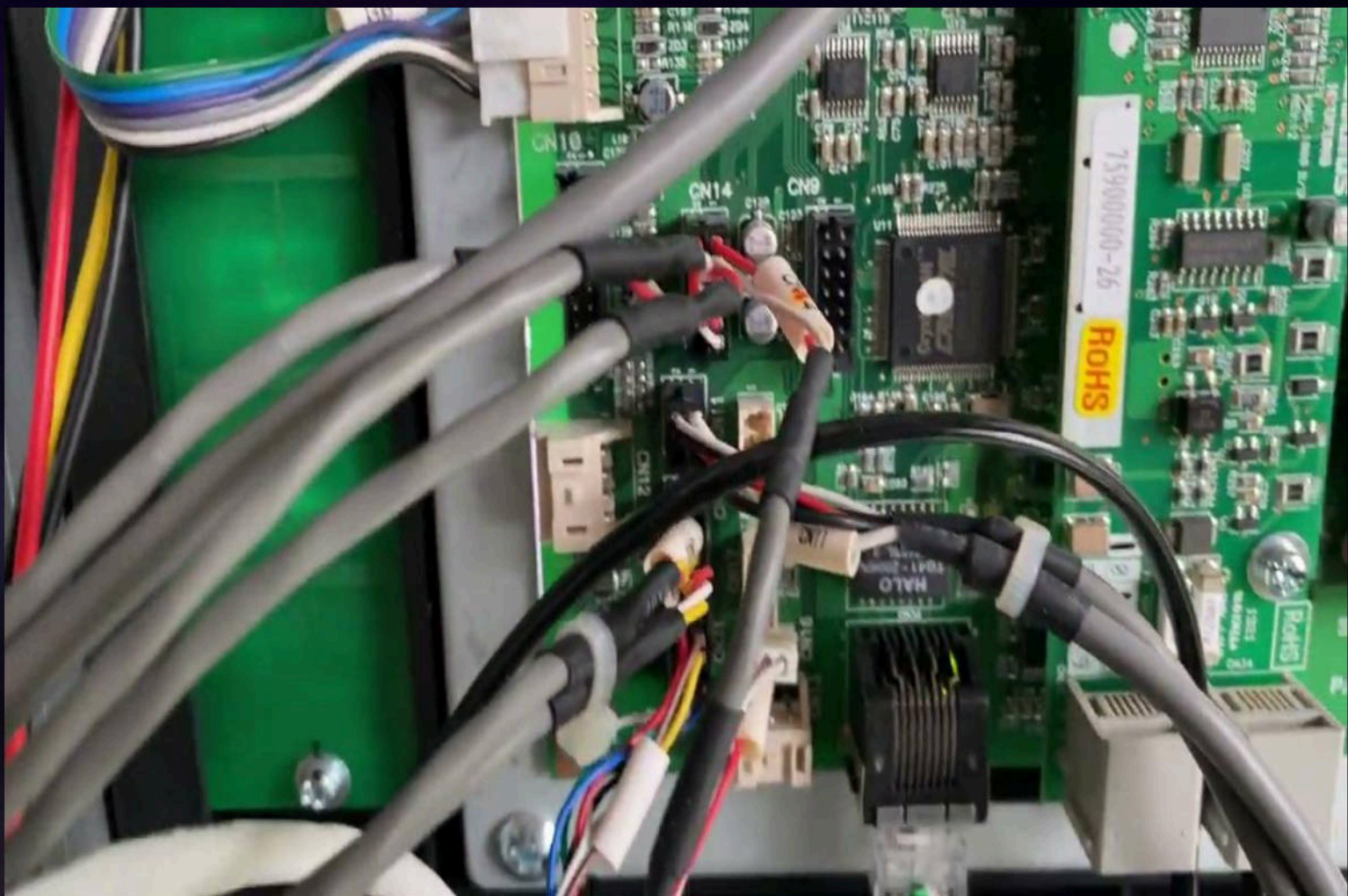
Add to Watchlist

100% buyer satisfaction

64 sold

Delivery in 4 days





Auth request with
card info and PIN



Internet

TLS
Encrypted Data
Triton Protocol

Approved
Denied



Payment Processing Host

Interbank

TLS
Encrypted Data
Protocol??

Approved
Denied



Card Issuing Bank



ATMDEPOT.COM

MORE MONEY IN YOUR POCKET





Allpoint

The logo for Allpoint features the word "Allpoint" in white, bold, sans-serif font on a green rectangular background. A small yellow icon of a person with arms raised is positioned above the letter 'i'.

NYCE
New York Cash Exchange

The logo for NYCE New York Cash Exchange features the word "NYCE" in large, bold, black, sans-serif font, tilted at an angle. Below it, the words "New York Cash Exchange" are written in a smaller, black, sans-serif font, also tilted at the same angle.

**MONEY
PASS**

The logo for MONEY PASS features the words "MONEY" and "PASS" in bold, sans-serif font, with "MONEY" in green and "PASS" in blue. The text is enclosed within a stylized, curved shape that resembles a swoosh or a partial circle, with the top part in green and the bottom part in blue.

pulse

The logo for pulse features the word "pulse" in a bold, italicized, black, sans-serif font. A thick orange horizontal bar is positioned below the letters "e" and "l".

PLUS

The logo for PLUS features the word "PLUS" in a bold, white, sans-serif font, set against a blue background. To the left of the text is a white graphic element consisting of a stylized triangle or arrow pointing upwards and to the right.

Cirrus

The logo for Cirrus features the word "Cirrus" in a bold, white, italicized, sans-serif font, set against a blue background. The background is composed of two overlapping circles, with the right circle having a horizontal striped pattern.

STAR

The logo for STAR features the word "STAR" in a bold, red, sans-serif font. Above the letter "A" is a red five-pointed star. A red swoosh or arc is positioned above the letters "S" and "T".

DEFCON

The logo for DEFCON features the word "DEFCON" in a stylized, multi-colored font, with each letter having a different color and a glowing effect. The letters are arranged in a slightly curved, horizontal line.

ATM Licensing



All data current as of Jul 14 at 1:45:26 PM ET, unless otherwise noted.

ENTITY PROFILE

LICENSES *

[Hide](#)

	Entities	Approved Licenses	Pending Licenses
Company	1	1	0
Branch	0	0	0
Individual - Sponsored	0	0	0

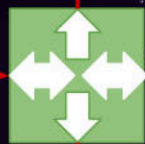
* Data current as of: Jul 14 at 1:00:20 PM ET

License Number	License Name	Status	Status Date	Original License Date	License Items	Adverse Status	Current Renewal Status	Renewed Through Year
	Minnesota Electronic Financial Terminal License	Approved	3/12/2021	3/12/2021	1	No		2021





Outbound
from ATM



Inbound
to ATM



PPH



Applications ▾ Places ▾ Wireshark ▾ Tue 02/07

Capturing from eth0 and eth1

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Filter: eth == 192.168.20.100 or eth == 192.168.10.100

No.	Time	Source	Destination	Protocol	Length	Info
100	99.73861228	192.168.20.100	ata.columbusdata...	TCP	60	80 49178 → 50117 [ACK] Seq=1 Win=2160 Len=0 MSS=1000
101	99.81504328	192.168.20.100	ata.columbusdata...	TCP	60	80 49178 → 50117 [ACK] Seq=1 Win=2160 Len=0
102	99.87702482	192.168.20.100	ata.columbusdata...	TLSv1.2	871	Client Hello
103	99.72502202	ata.columbusdata.info	192.168.20.100	TCP	60	50117 → 49178 [ACK] Seq=1 Ack=510 Win=1517 Len=0
104	99.73890322	ata.columbusdata.info	192.168.20.100	TLSv1.2	1514	Server Hello
105	99.74015454	ata.columbusdata.info	192.168.20.100	TCP	1514	50117 → 49178 [ACK] Seq=140 Ack=510 Win=1517 Len=14
106	99.74245124	192.168.20.100	ata.columbusdata...	TCP	60	49178 → 50117 [ACK] Seq=510 Ack=2021 Win=2160 Len=0
107	99.74142922	ata.columbusdata.info	192.168.20.100	TCP	1514	50117 → 49178 [ACK] Seq=2021 Ack=510 Win=1517 Len=14
108	99.74246041	ata.columbusdata.info	192.168.20.100	TLSv1.2	1170	Certificate, Server Key Exchange, Server Hello Done
109	99.74365072	192.168.20.100	ata.columbusdata...	TCP	60	49178 → 50117 [ACK] Seq=510 Ack=5506 Win=2160 Len=0
110	100.246099	192.168.20.100	ata.columbusdata...	TLSv1.2	180	Client Key Exchange, Change Cipher Spec, Encrypted Handshake
111	100.348487	ata.columbusdata.info	192.168.20.100	TCP	60	50117 → 49178 [ACK] Seq=5506 Ack=644 Win=13143 Len=0
112	100.346018	ata.columbusdata.info	192.168.20.100	TLSv1.2	150	Change Cipher Spec, Encrypted Handshake Message
113	100.436725	ata.columbusdata.info	192.168.20.100	TCP	60	TCP ACK Seq=644 Win=0 Seq=140 Ack=5506 Len=0
114	100.436725	192.168.20.100	ata.columbusdata...	TLSv1.2	64	TCP Seq=140 Win=0 Seq=140 Ack=5506 Len=0
115	100.461082	192.168.20.100	ata.columbusdata...	TCP	60	TCP ACK Seq=644 Win=0 Seq=140 Ack=5506 Len=0
116	100.540134	ata.columbusdata.info	192.168.20.100	TLSv1.2	60	TCP Seq=140 Win=0 Seq=140 Ack=5506 Len=0

Frame 78: 135 bytes on wire (1040 bits), 131 bytes captured (1040 bits) on interface 1

Ethernet II, Src: Cisco-L3_eb:8f:78 (48:1d:7e:eb:8f:78), Dst: Foryangn_A1:a8:64 (80:08:0c:a1:a8:64)

Internet Protocol Version 4, Src: 192.168.20.1, Dst: 192.168.10.100 (192.168.10.100)

User Datagram Protocol, Src Port: domain (53), Dst Port: 50038 (50038)

Domain Name System (response)

```

0000  00 00 70 41 40 00 10 7e 00 5f 78 00 00 40 00  |A 0 70 41 40 00 10 7e 00 5f 78 00 00 40 00|
0010  00 75 40 07 40 00 3f 11 73 3a c0 00 50 01 c0 09  |07 40 07 40 00 3f 11 73 3a c0 00 50 01 c0 09|
0020  00 00 00 00 00 c0 00 00 00 00 00 00 00 00 00 00  |00 00 00 00 00 c0 00 00 00 00 00 00 00 00 00|
0030  00 07 00 00 00 00 00 41 34 40 00 43 47 4c 50 43  |07 00 00 00 00 00 00 41 34 40 00 43 47 4c 50 43|
0040  42 50 12 44 41 54 47 63 4e 40 54 00 00 01 00 01  |42 50 12 44 41 54 47 63 4e 40 54 00 00 01 00 01|
0050  c0 dc 40 00 00 00 00 00 01 1f 00 17 00 01 34 0d  |c0 dc 40 00 00 00 00 00 01 1f 00 17 00 01 34 0d|
0060  00 03 0f 00 75 0d 62 05 13 04 01 74 01 04 00 00  |00 03 0f 00 75 0d 62 05 13 04 01 74 01 04 00 00|
    columbusdata.in
  
```

eth0 and eth1: Live capture in progress

Packets: 222 | Displayed: 47 (21.2%)

Profile: Default



HYOSUNG

TCP/IP TYPE

TYPE

STANDARD

SSL/TLS

ENABLE

SSL/TLS VERSION

TLS v1.2

SSL/TLS CERT.

ENABLE

When SSL/TLS is enabled, certificate validation will be enabled by default. It can be disabled but strongly recommend using certificate validation for maximum transaction security. For this to work properly you must load the appropriate certificate for your host processor.

CANCEL TO RETURN

VISA FRAMED

STANDARD

ACK
CONTROLLED

SSL/TLS
EN/DISABLE

SSL/TLS
VERSION

SSL/TLS CERT.
EN/DISABLE

DOWNLOAD CERT.
FROM USB

DOWNLOAD CERT.
FROM USB

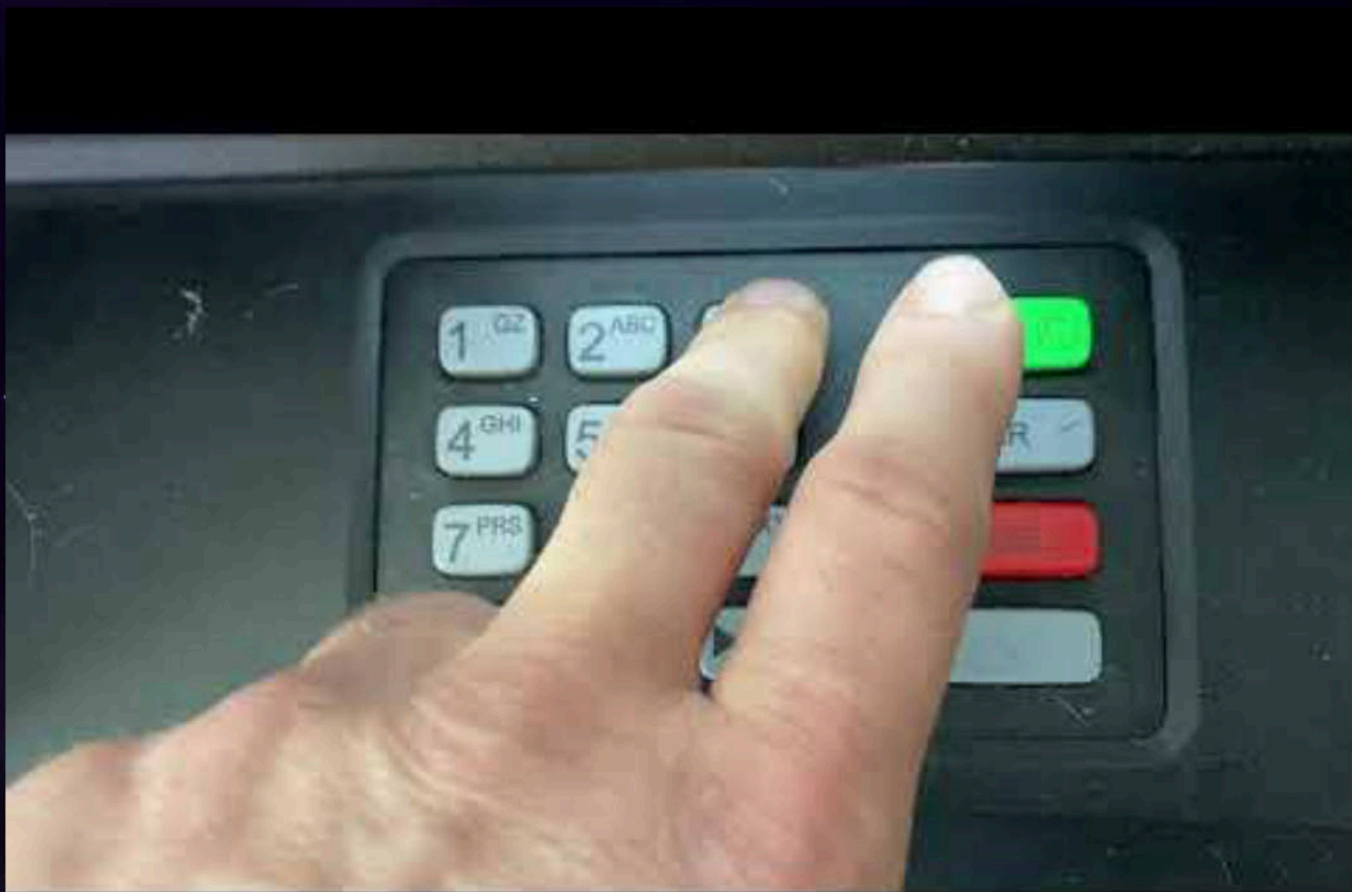




OUT OF SERVICE

ATM is temporarily out of service.

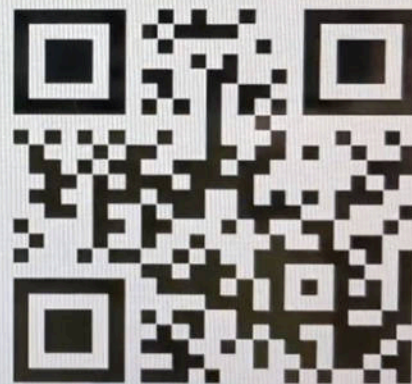
ERROR CODE : 0-FFFFFF(FF)



HYOSUNG

ENTER PASSWORD

1	2	3	ENTER
4	5	6	CLEAR
7	8	9	CANCEL
.	0	,	



CANCEL TO RETURN

DEFAULT PASSWORDS:

When machine passwords are reset to default, they should be as follows:

HYOSUNG:

MASTER: 555555

SERVICE: 222222

OPERATOR: 111111

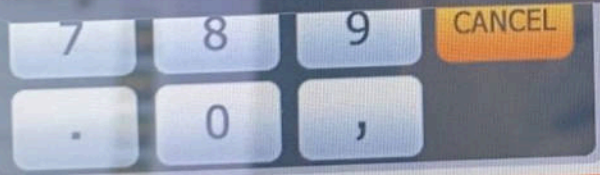
<https://bit.ly/3z4Azm2>



HYOSUNG

ENTER PASSWORD

Invalid Password



CANCEL TO RETURN



C:\Users\admin\Downloads\ATM\VUS_WINCE60_V060205_OS_BL_SDCard

File

Home

Share

View



<< ATM > VU... >



Search VUS_WINCE60_V060205_OS_BL_SDCard

> Quick access

> Desktop

Name

Date modified

Type

Size



update

7/4/2021 8:19 AM

File folder



EBOOT21s.bin

7/28/2020 1:56 PM

BIN File

235 KB



EBOOTA8.bin

7/28/2020 1:56 PM

BIN File

381 KB



EBOOTA8S.bin

7/28/2020 1:56 PM

BIN File

385 KB



nhA8snk.bin

7/28/2020 1:56 PM

BIN File

40,961 KB



nhv21snk.bin

7/28/2020 1:56 PM

BIN File

19,457 KB



SDUPDATE.tns

7/28/2020 1:56 PM

TNS File

1 KB



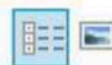
stplda8s.bin

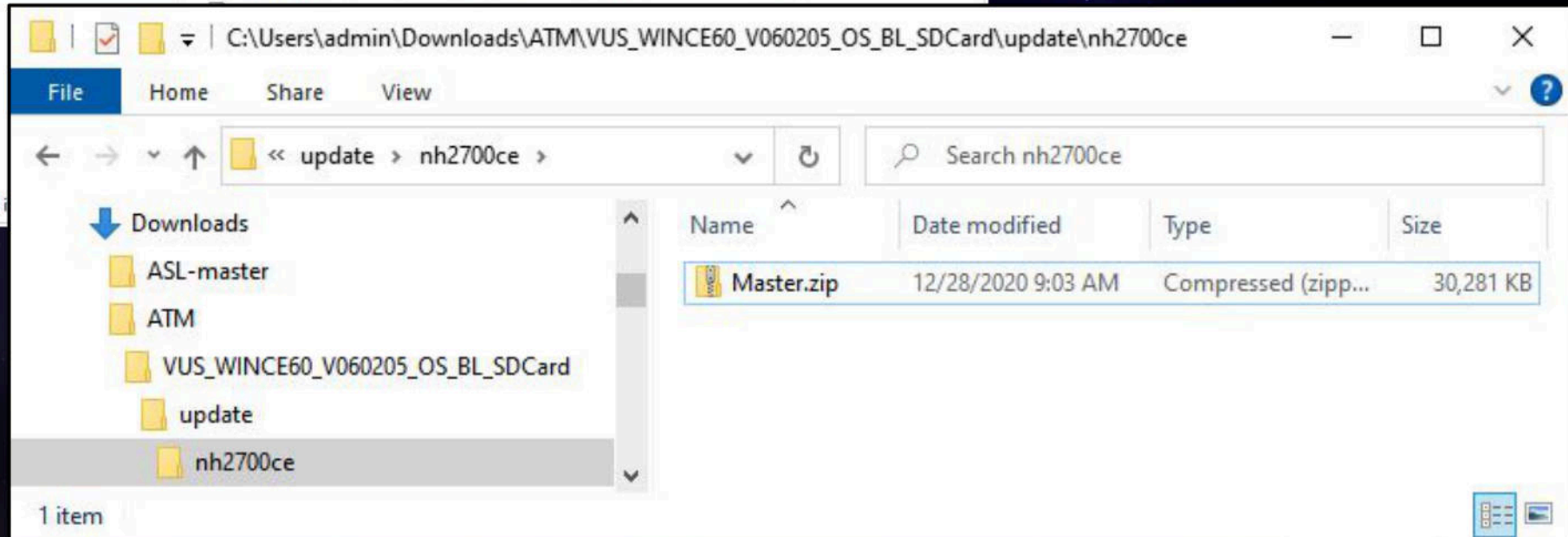
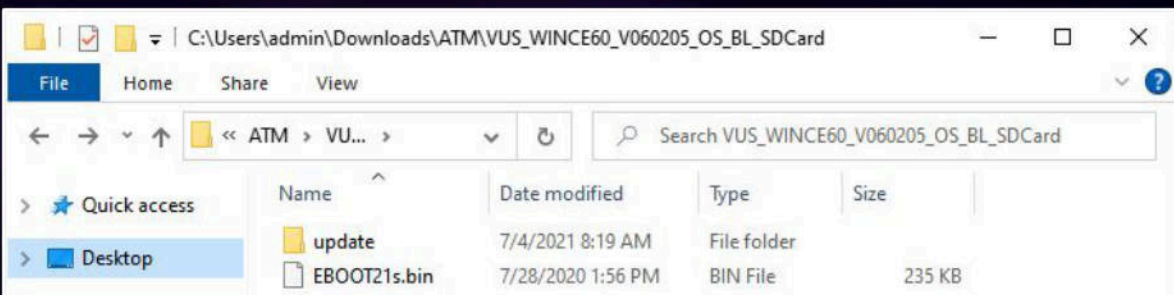
7/28/2020 1:56 PM

BIN File

5 KB

8 items





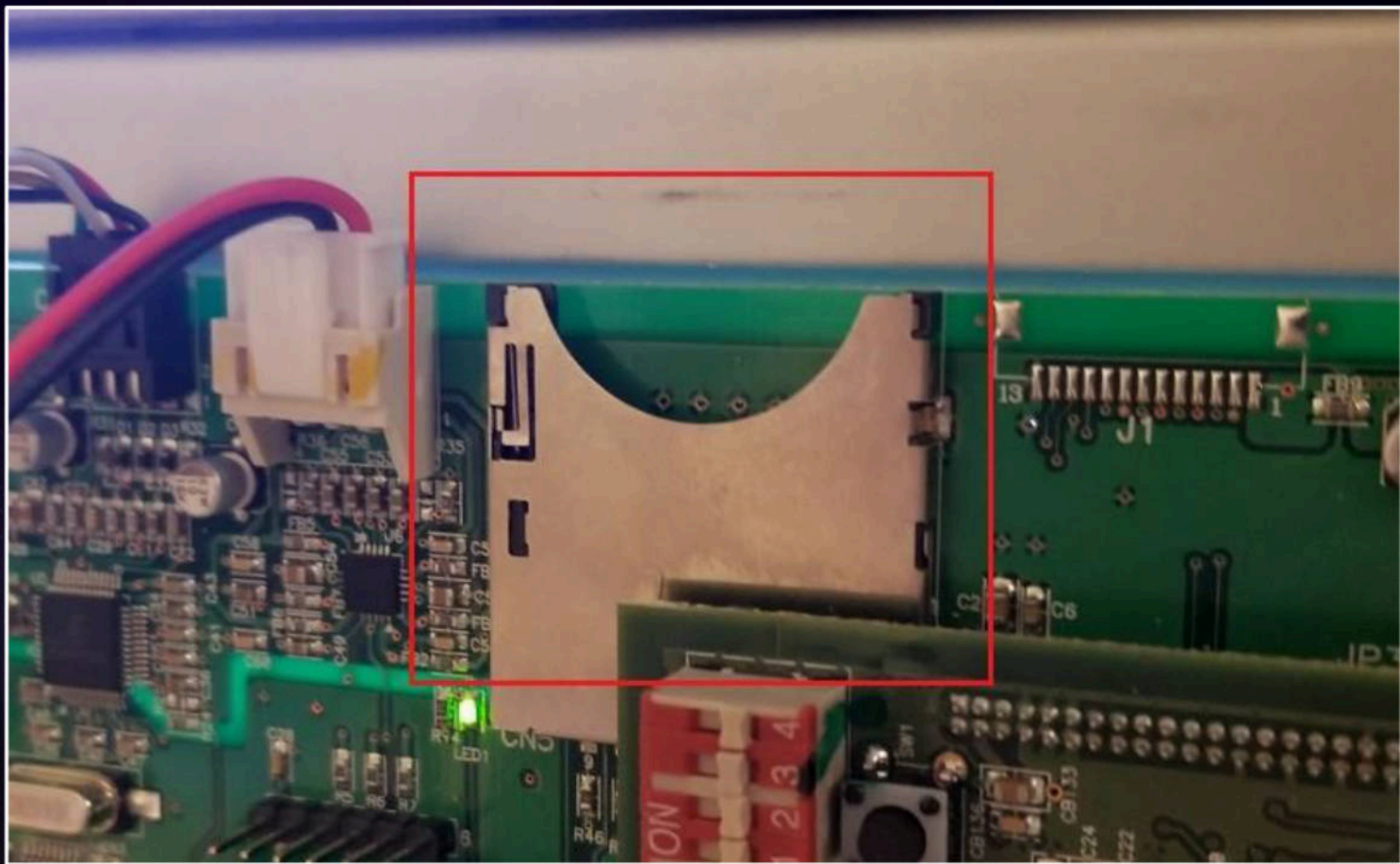
Master >

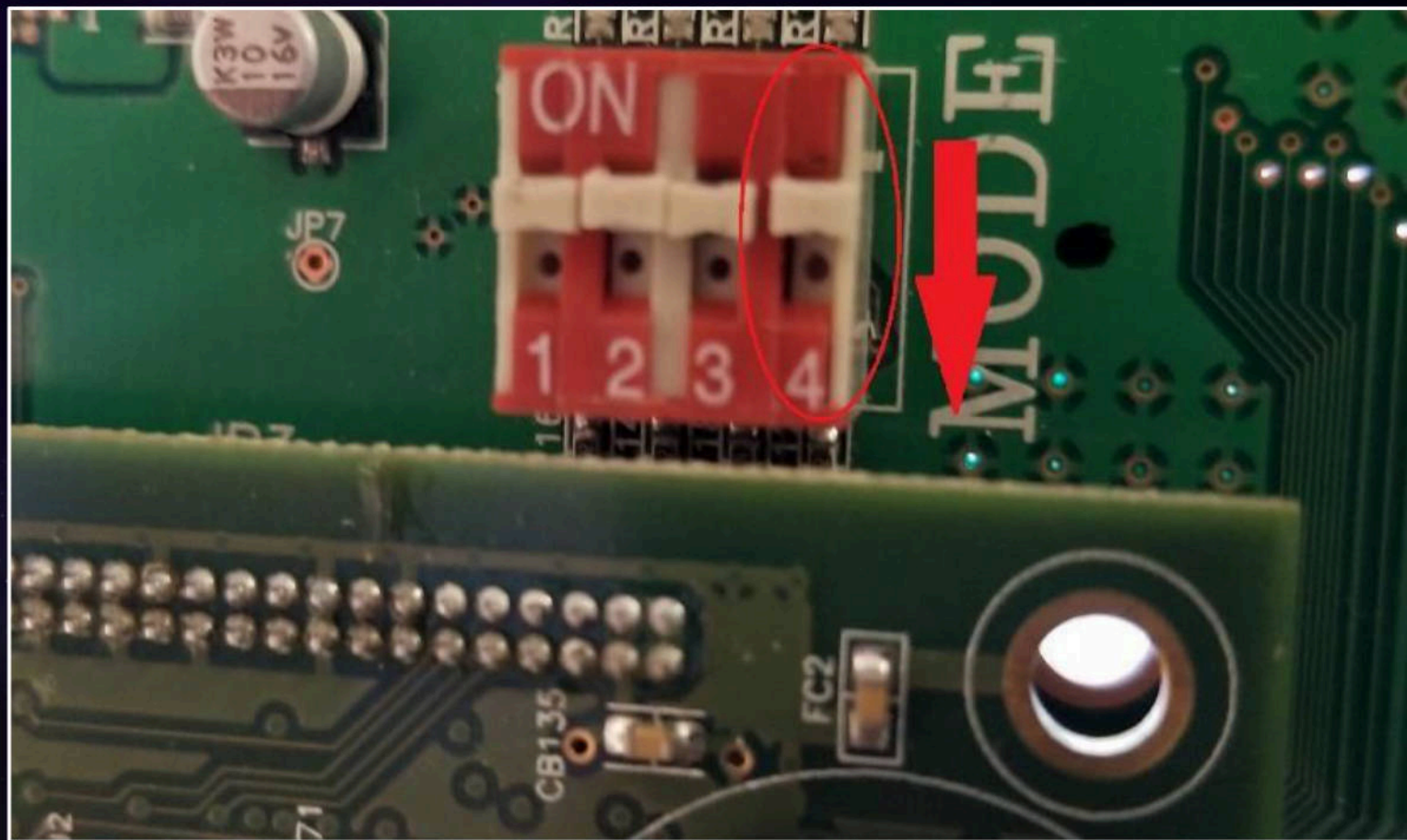


Search Master

Name	Date modified	Type	Size
Data	7/4/2021 8:55 AM	File folder	
EMVINI	7/4/2021 8:55 AM	File folder	
ERRORCODE	7/4/2021 8:55 AM	File folder	
SCREEN	7/4/2021 8:55 AM	File folder	
TTS_DB	7/4/2021 8:55 AM	File folder	
WAVE	7/4/2021 8:55 AM	File folder	
2420-03D.DWL	12/15/2020 1:40 PM	DWL File	91 KB
3898-02C.DWL	12/15/2020 1:40 PM	DWL File	101 KB
4707-02G.DWL	12/15/2020 1:40 PM	DWL File	92 KB
ADACtrl.dll	12/15/2020 1:43 PM	Application exten...	62 KB
WatchDog.exe	12/15/2020 1:43 PM	Application	39 KB
WinAtm.exe	12/15/2020 1:43 PM	Application	1,240 KB
WinATMClear.bat	12/15/2020 1:40 PM	Windows Batch File	1 KB







NAUTILUS HYOSUNG (Ver : 086.22.15)

***** DIAGNOSTIC MODE *****

<<< SERIAL

SD CARD >>>

<<< LAN(MAC)

GUIDE LIGHT >>>



***** DIAGNOSTIC MODE *****



Software Update

Check update files

[Extracting...] 14% - 25/ 179

Default Backs2ipini

Please wait until ATM reboot





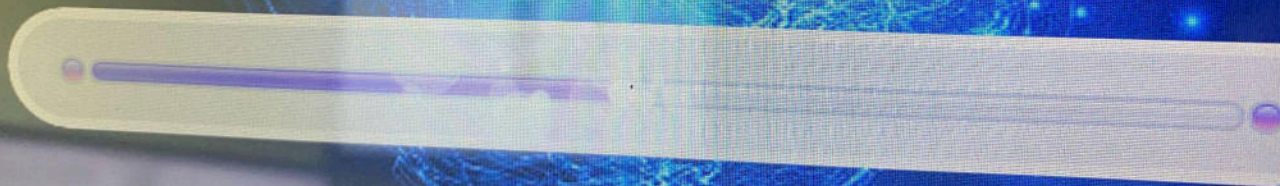
OUT OF SERVICE

ATM is temporarily out of service.

ERROR CODE : 0-FFFFFF(FF)



Please wait while initializing...



CLEAR

LEFT

RIGHT

CLEAR

CLEAR

CANCEL





OUT OF SERVICE

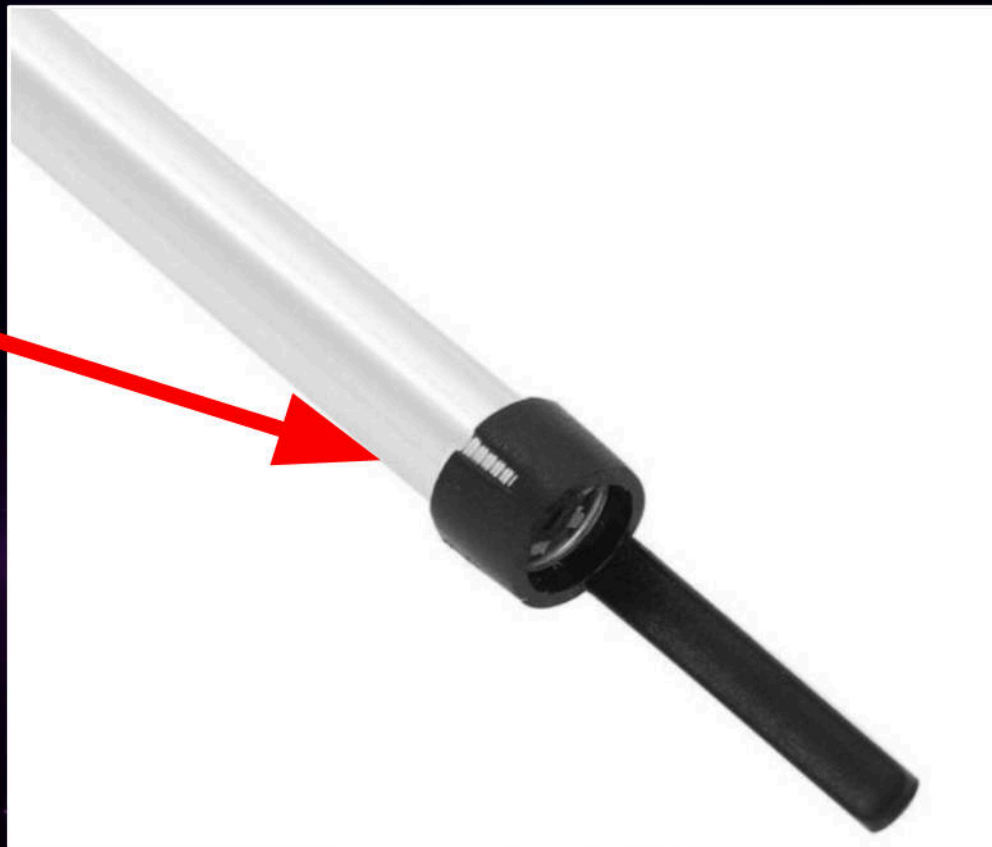
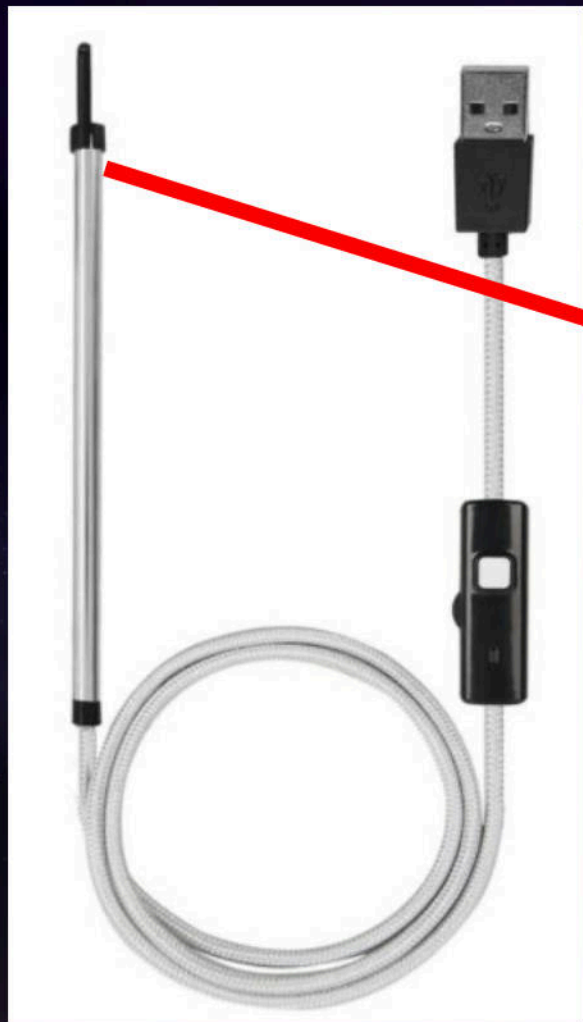
ATM is temporarily out of service.

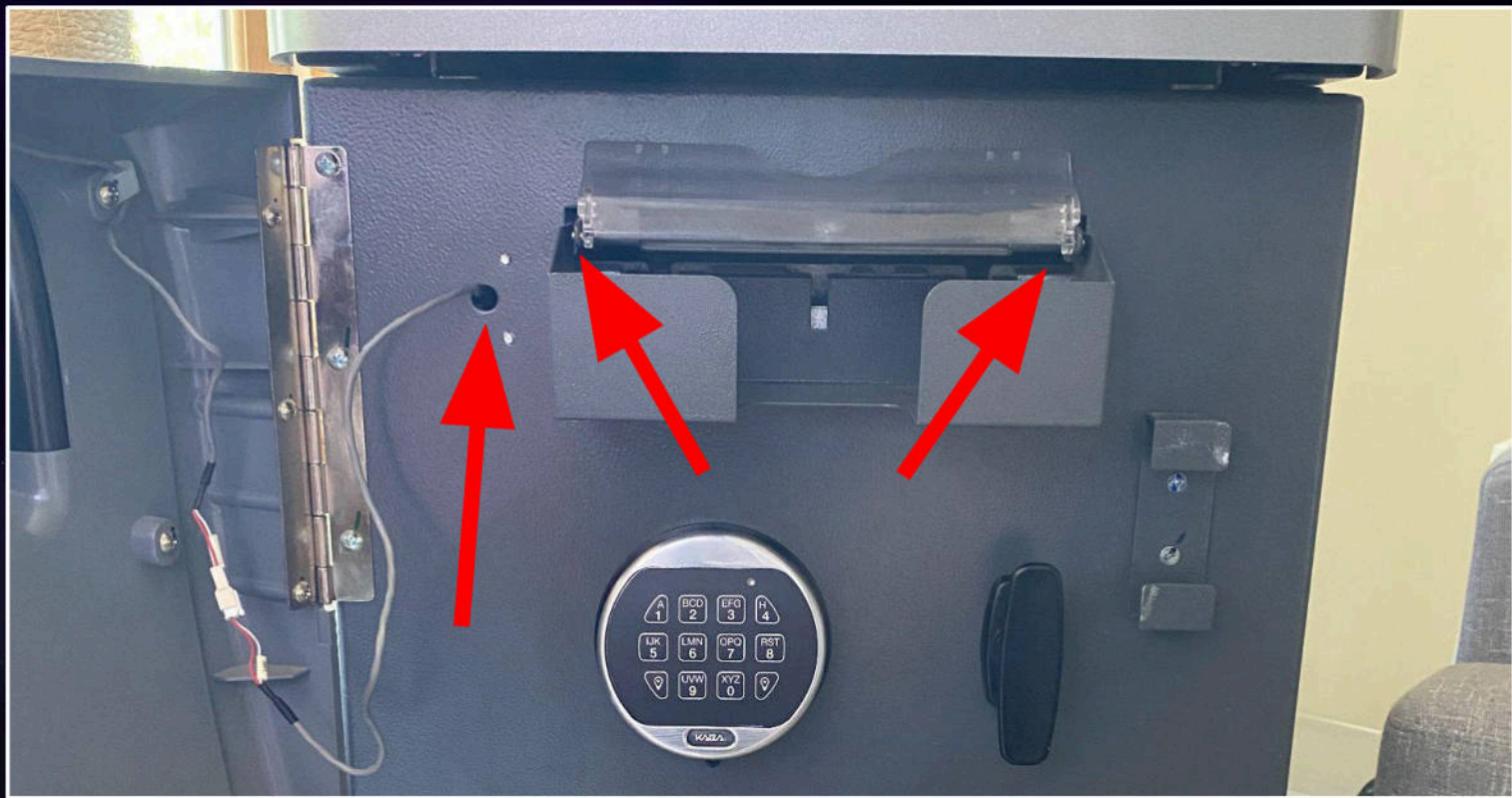
ERROR CODE : 0-FFFFFF(FF)





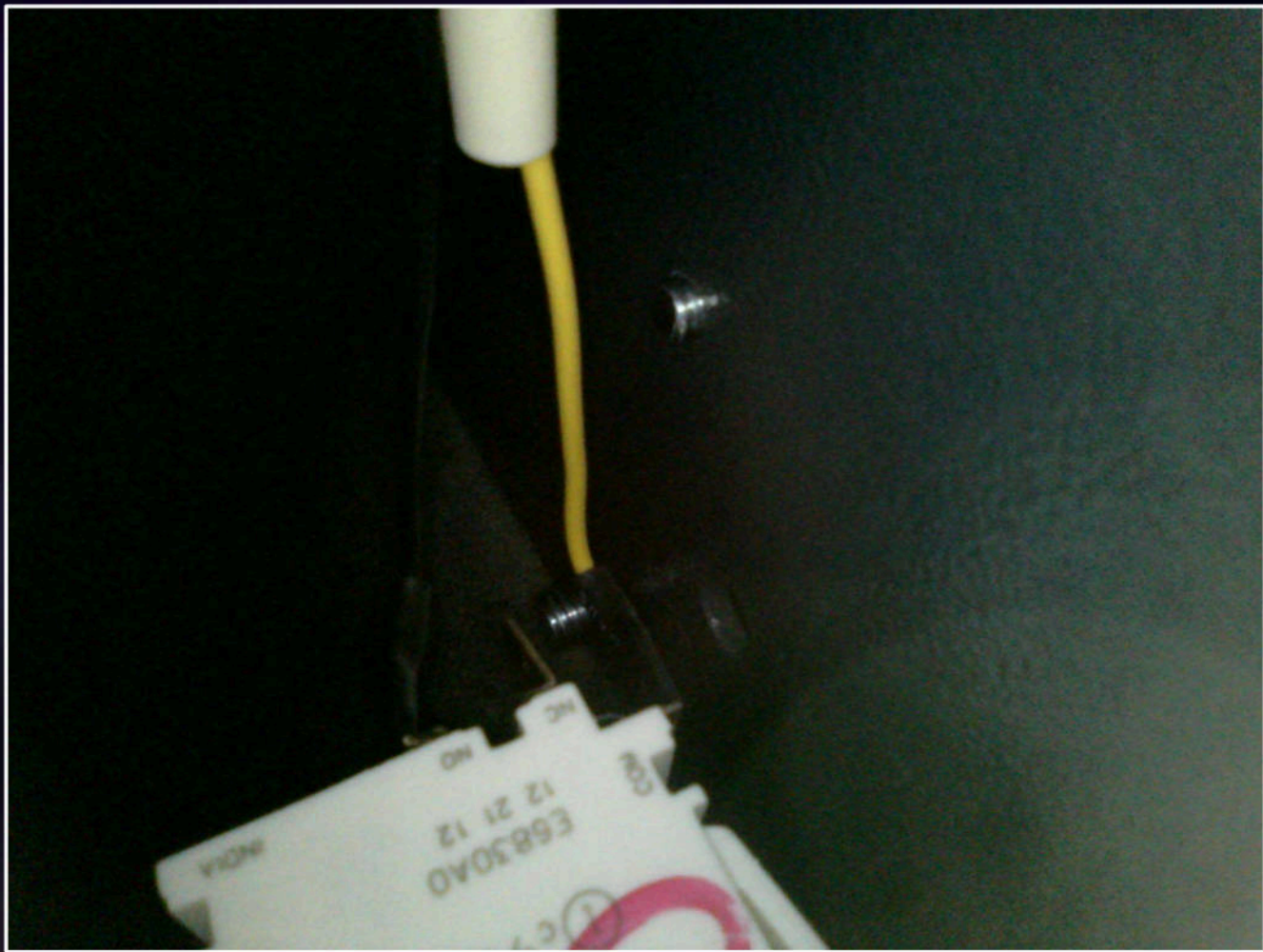




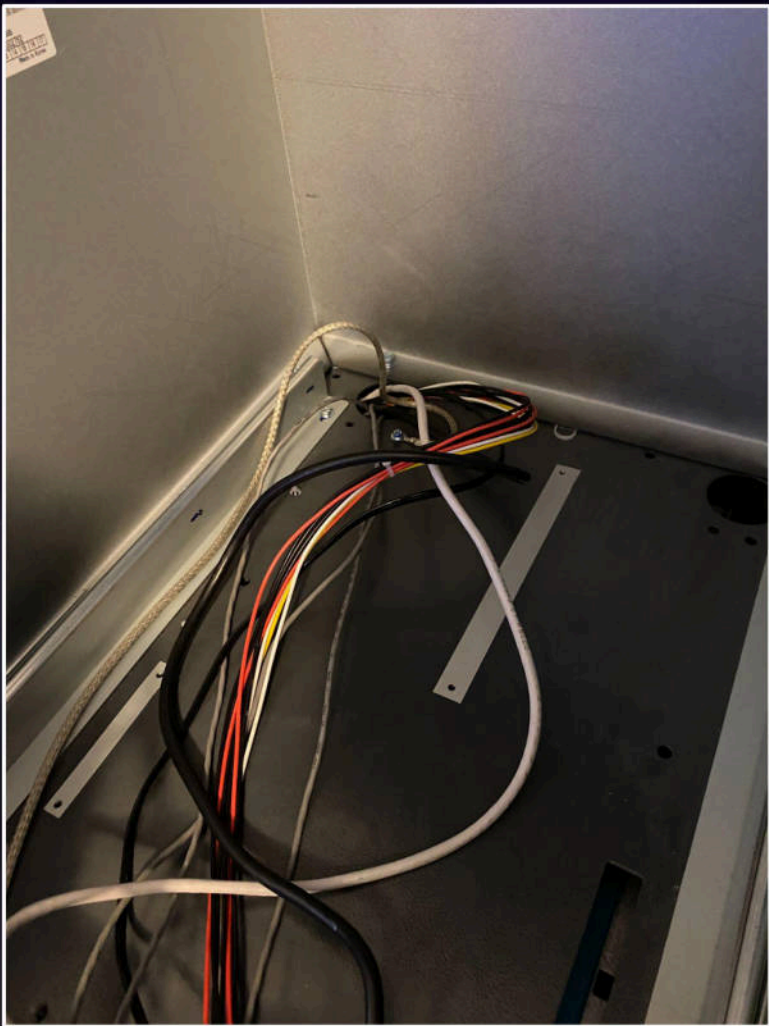


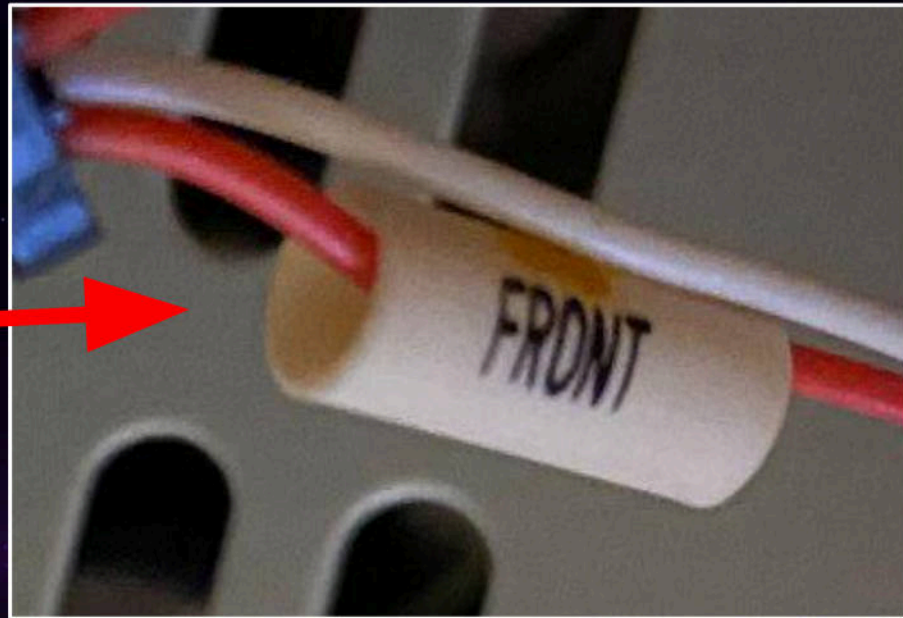
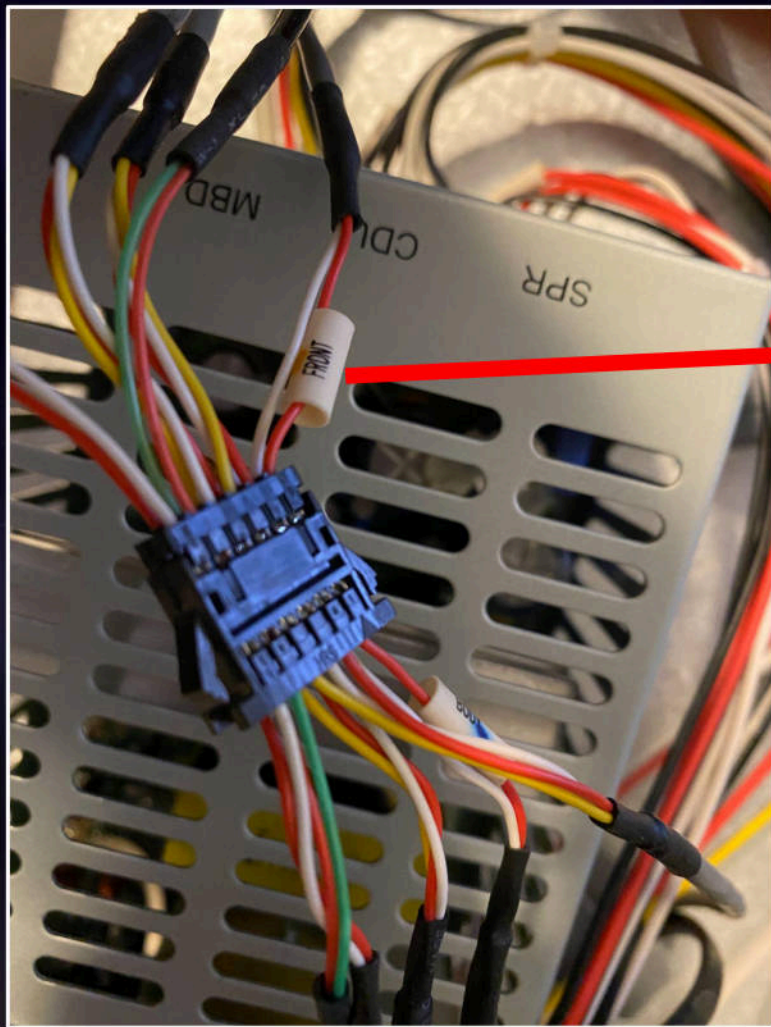


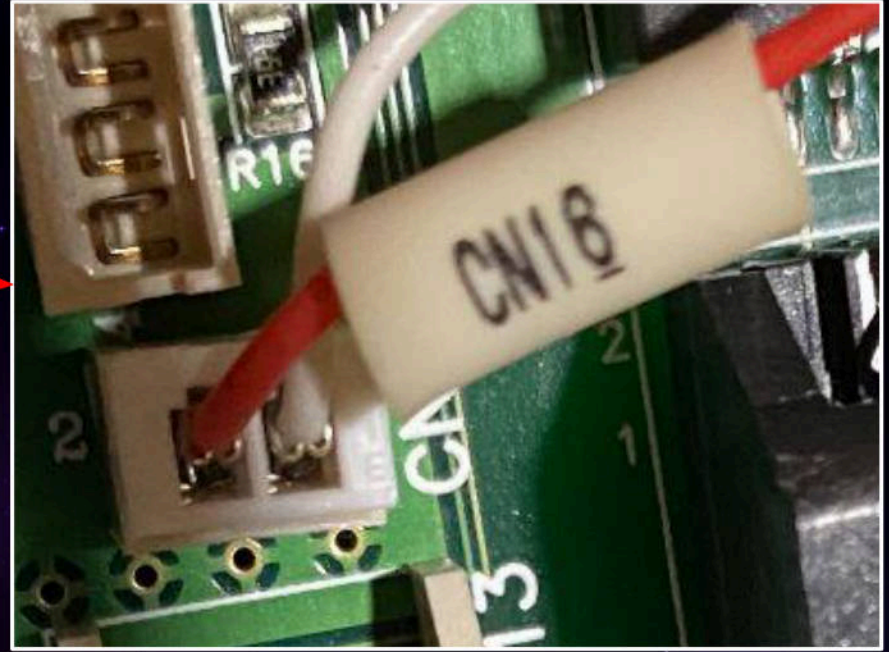
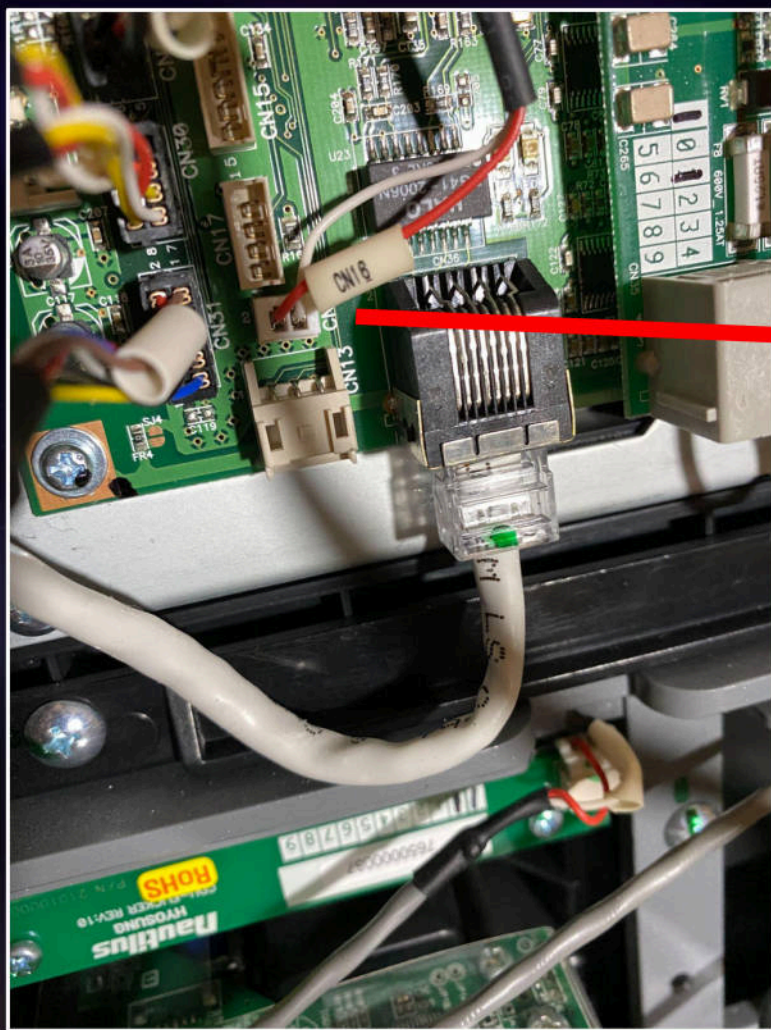






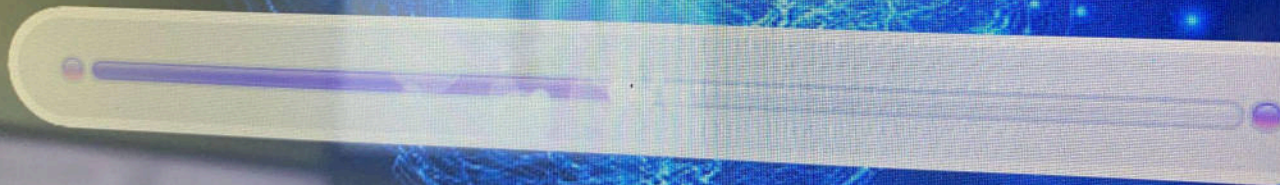








Please wait while initializing...



CLEAR

LEFT

RIGHT

CLEAR

CLEAR

CANCEL



? Reset Master Password

To reset master password
and clear EPP,
press 'YES'. To exit, press 'NO'.
The terminal will be rebooted
after pressing Button.

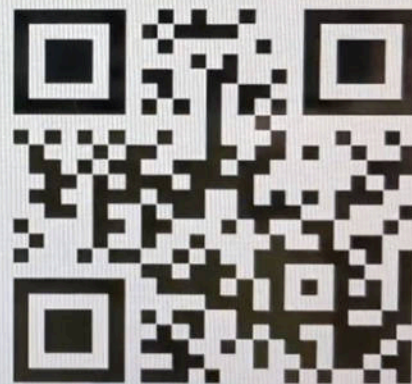
Yes

No

HYOSUNG

ENTER PASSWORD

1	2	3	ENTER
4	5	6	CLEAR
7	8	9	CANCEL
.	0	,	



CANCEL TO RETURN

HYOSUNG

Operation Program

01/24/2021 22

NI2700 USA

TERMINAL

00000000

SERIAL NO

0000000000

SETTLEMENT

JOURNAL

REPORT

DIAGNOSTICS

[] - [971A6(00)]

Denomination Not Set

RESET

Press "CLEAR" key
to reset terminal error

DEVICE STATUS

CARD



PINPAD



CDU



RECEIPT



RFID



CASH STATUS

CST 1



Empty
0(\$0)

CST 2



NO
--

CST 3



NO
--

CST 4



NO
--

REJECT

Normal
0

3 QUICK
REPLENISH

4 QUICK
CONFIGURATION

1 IN
SERVICE

2

PROTOCOL

STAND

NETWORK

TC

AP VERSION

V06.1

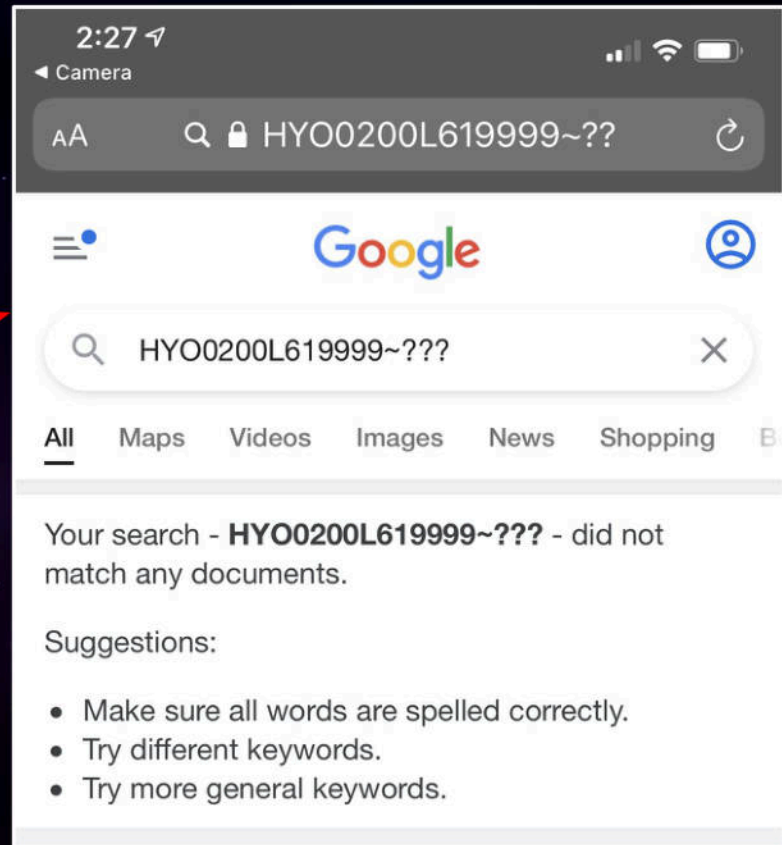
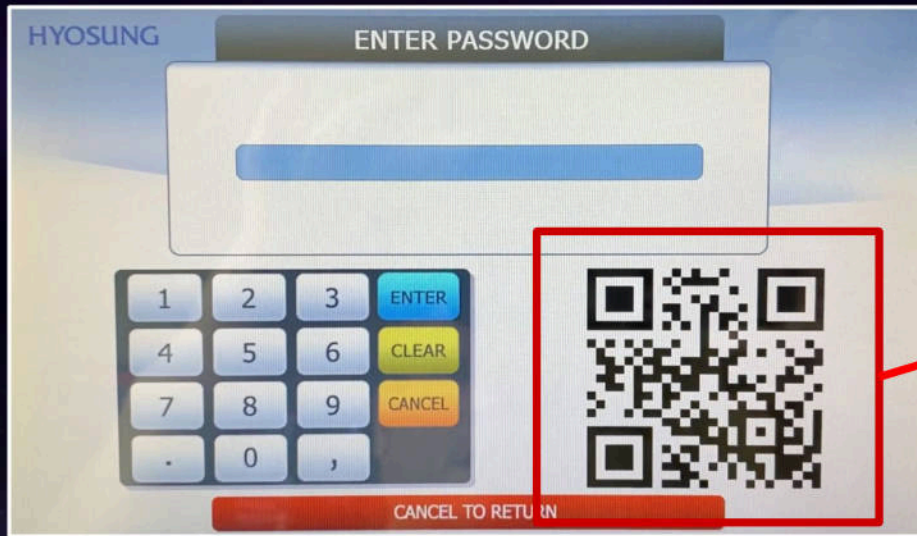
CUSTOMER
SETUP

SYSTEM SETUP

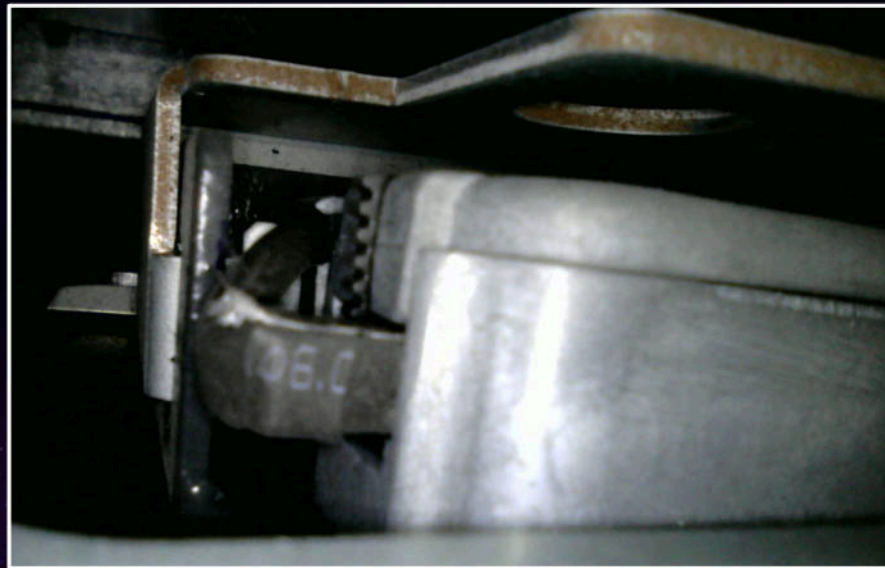
HOST SETUP

TRANSACTION
SETUP

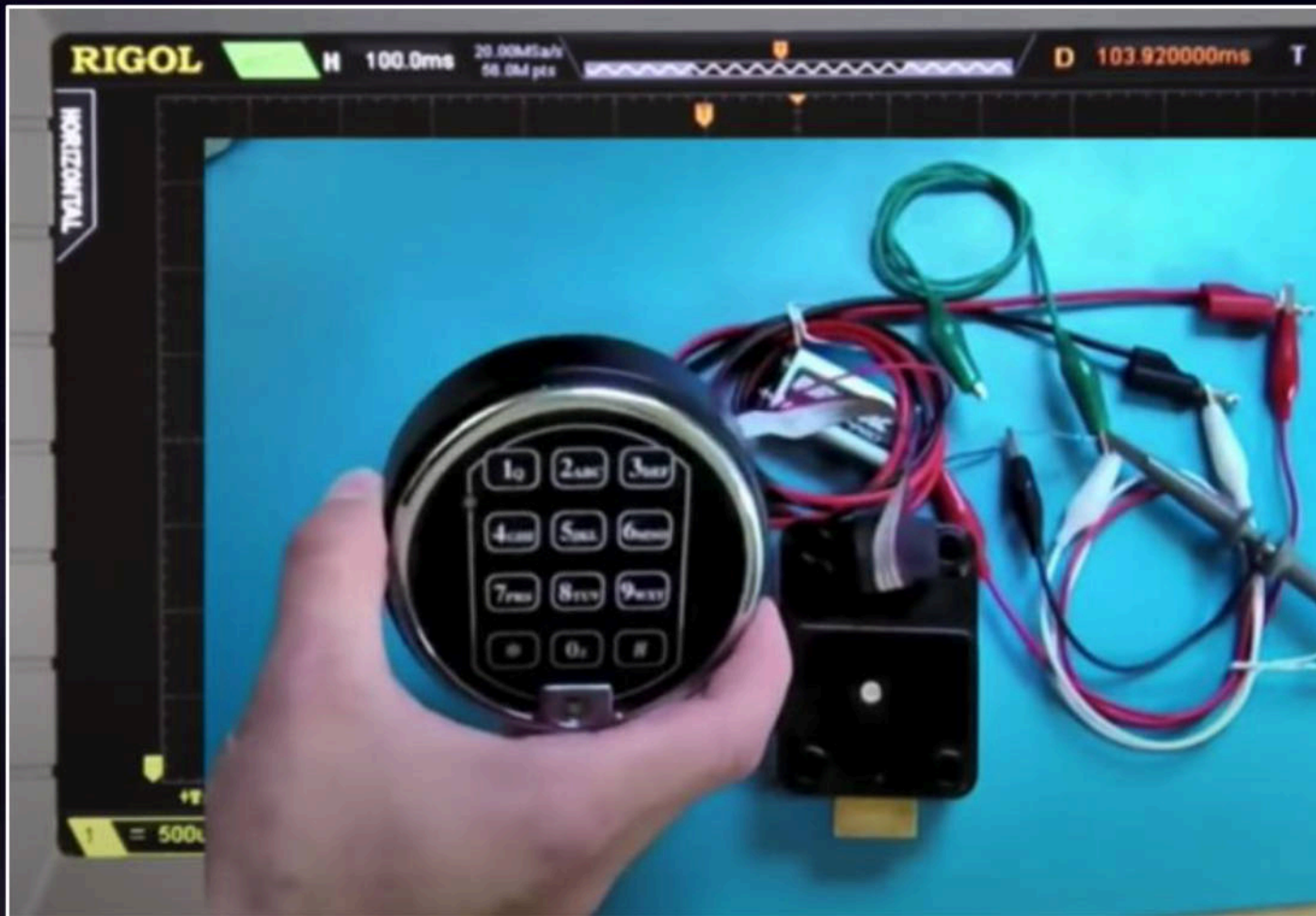
DEFCON











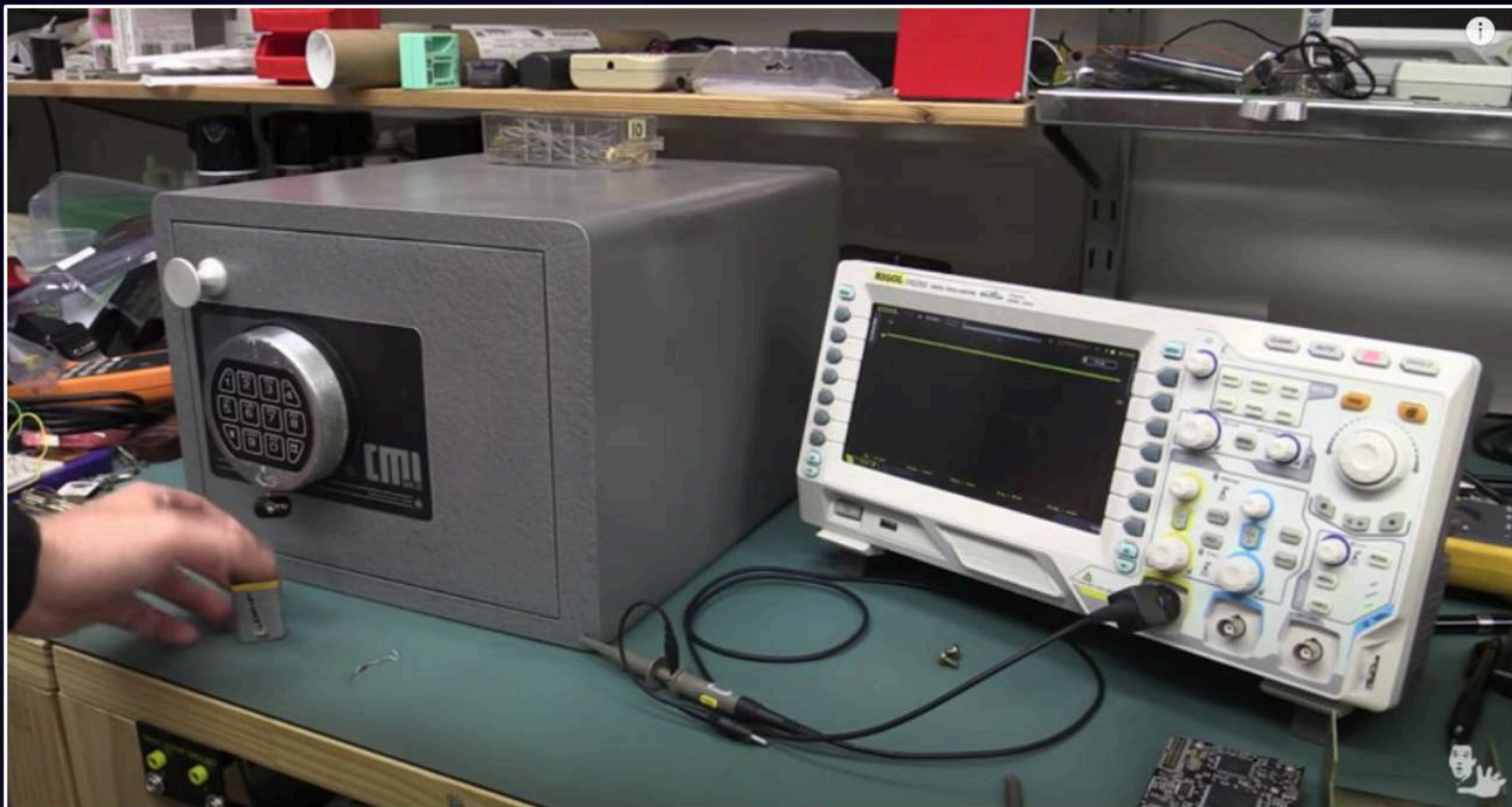
<https://www.youtube.com/watch?v=IXFpCV646E0&t=338s>





<https://www.youtube.com/watch?v=IXFpCV646E0>





<https://www.youtube.com/watch?v=HxQUKAjq-7w>





1830CTS

Updates & Warranty are void if cover is removed.

Analyzing...

045288457882309093747
653552379967366269198
1897973480549113626590
086655168795748183173
04221772858277514816
263665335481507311858



Battery: 18.24V

THE LITTLE
BB
BLACK BOX

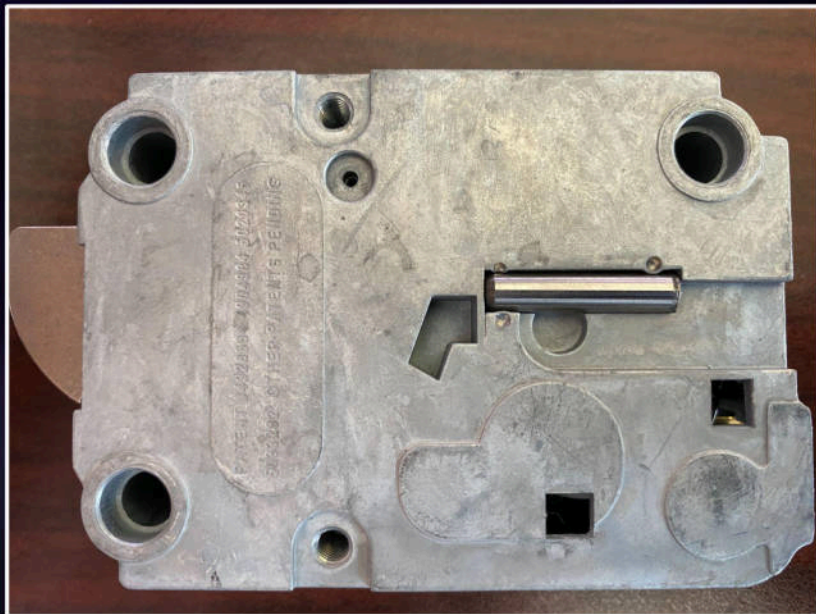
800.654.0637
www.lockmasters.com

DEFCON



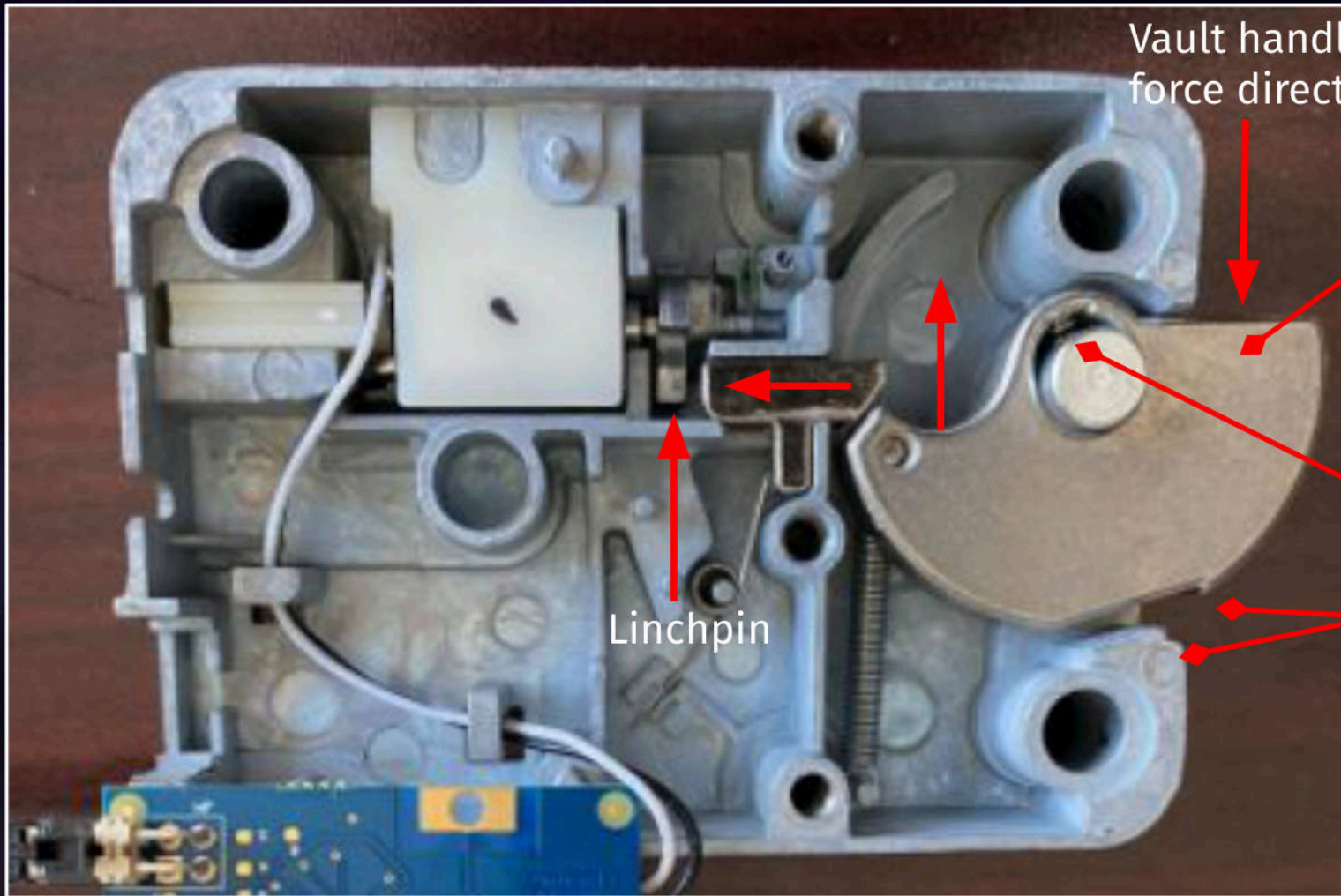
\$3,095.00





To Keypad





Vault handle
force direction

Lock bolt

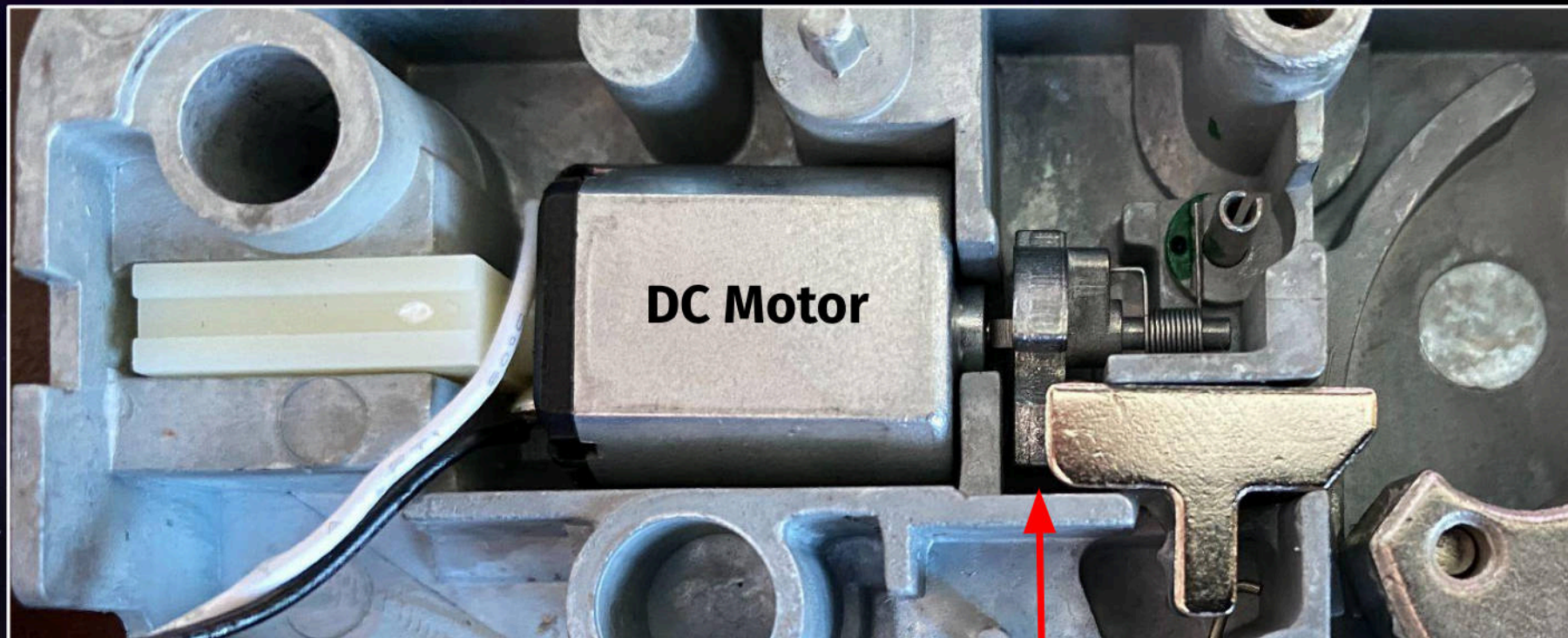
Forced down
by front
handle
rotation

Spring

Anti-force
notch

Linchpin



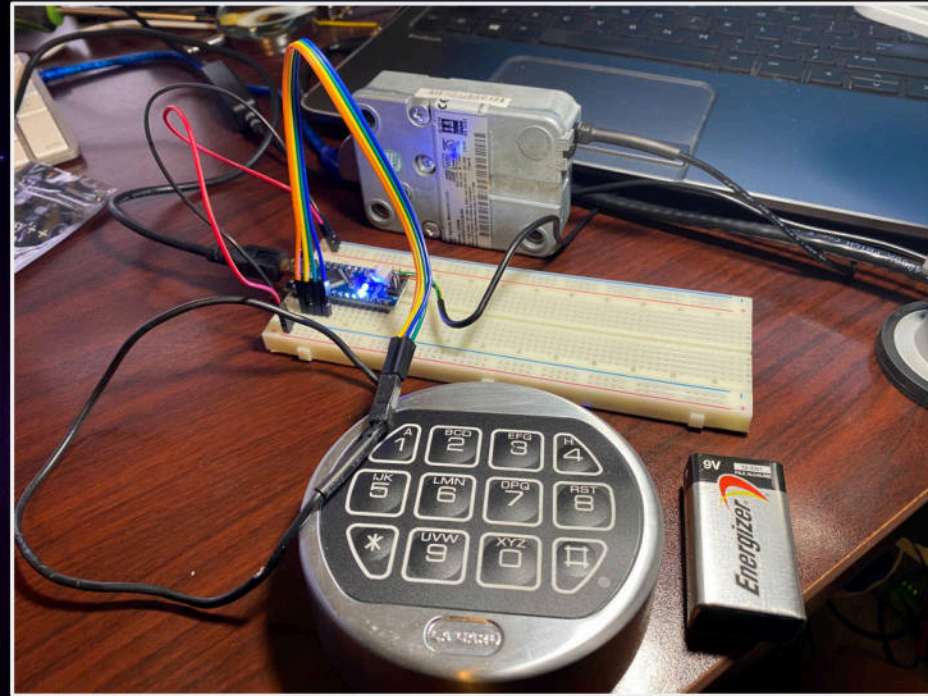


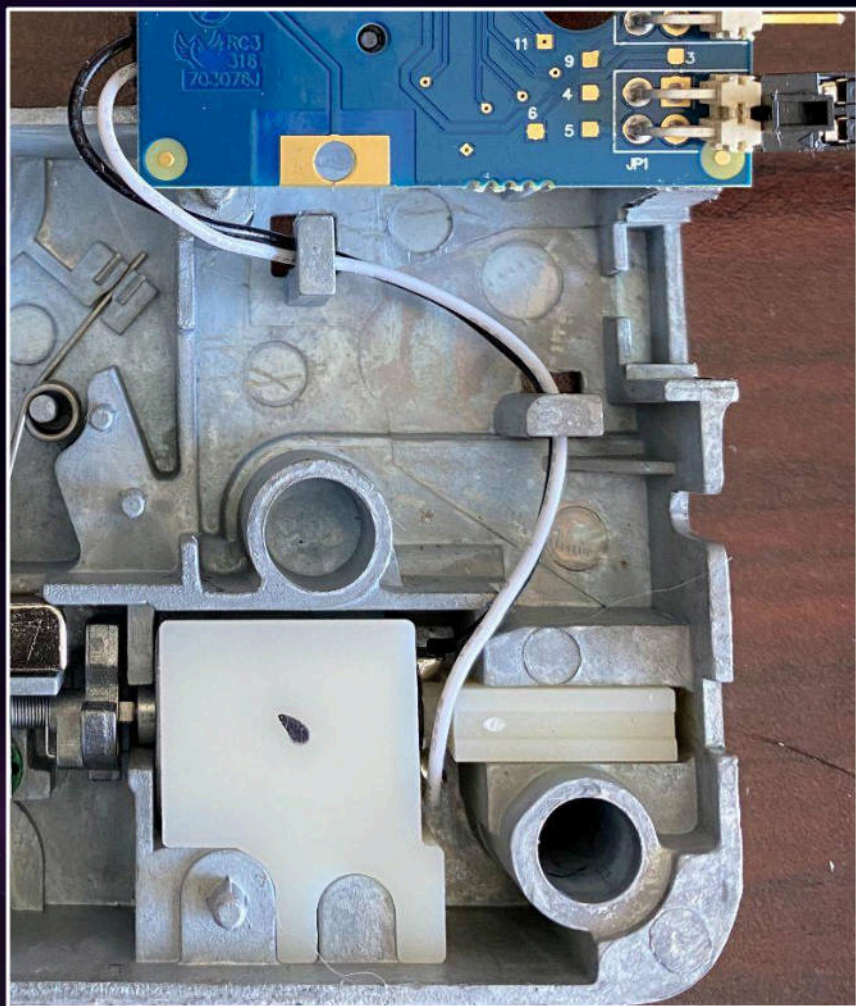
DC Motor

Linchpin

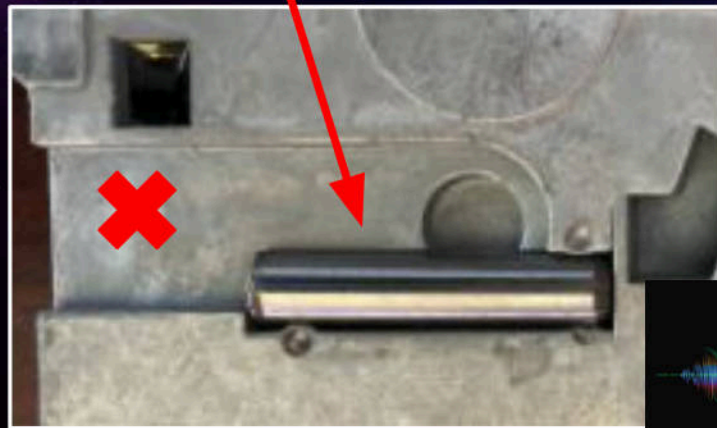
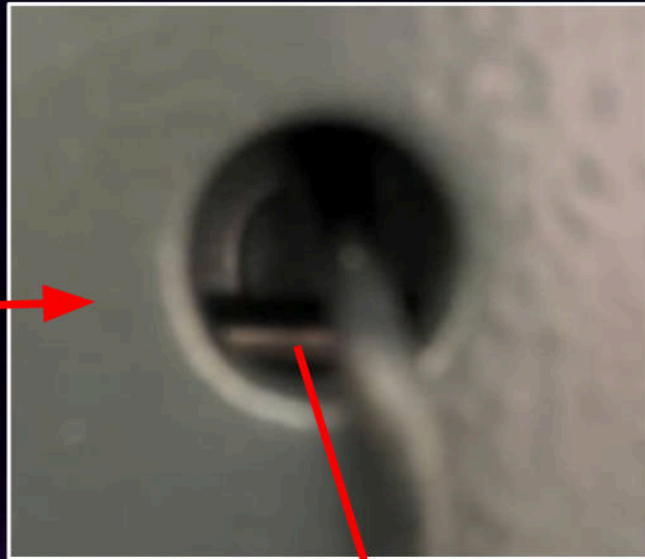
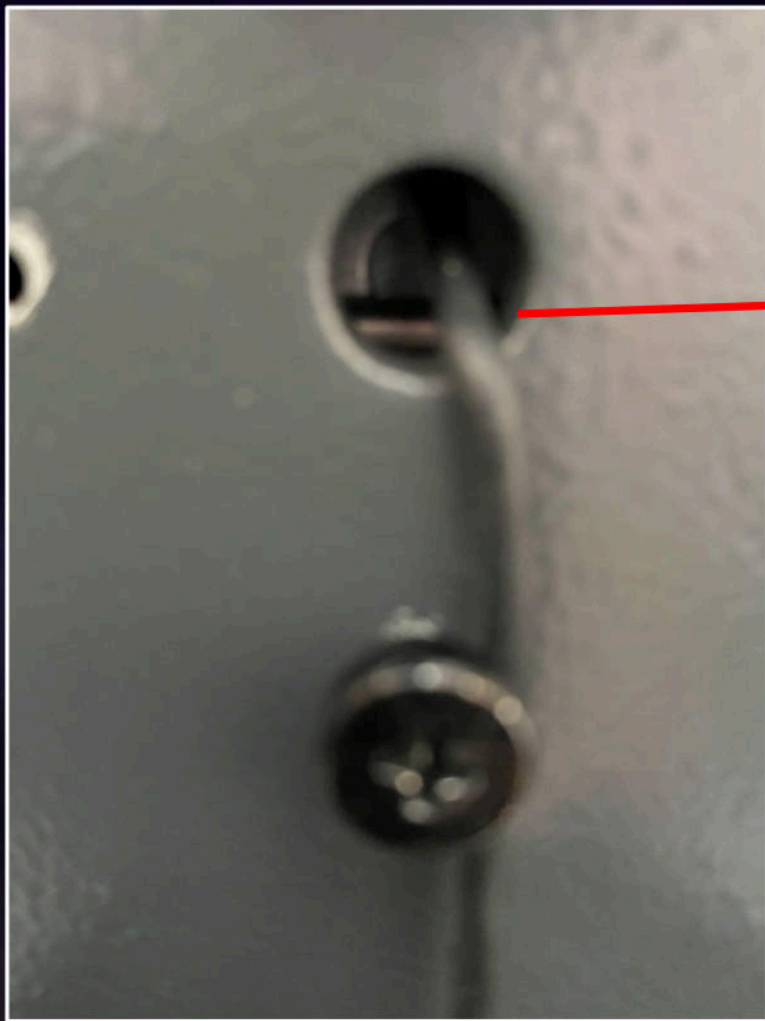








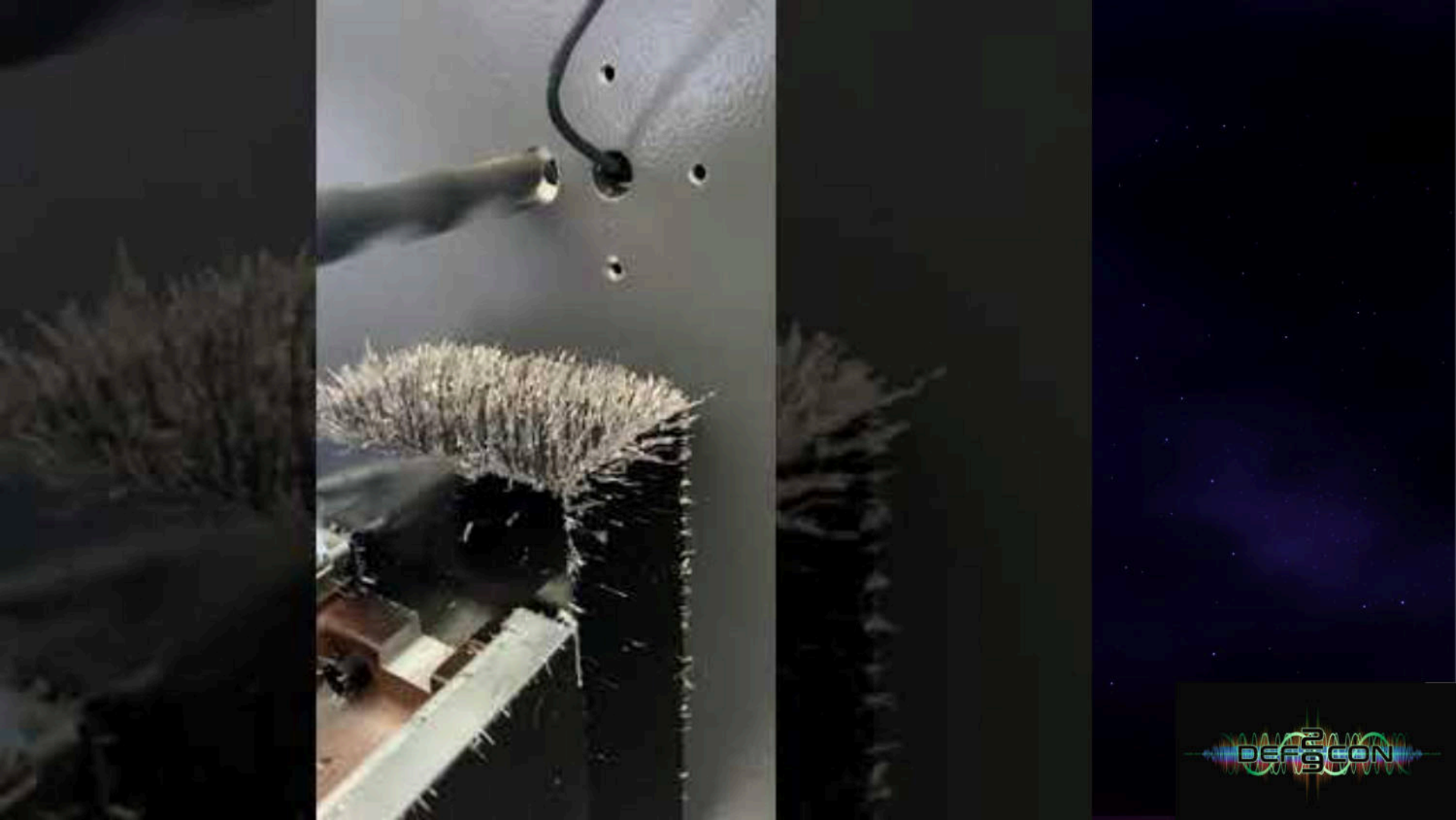


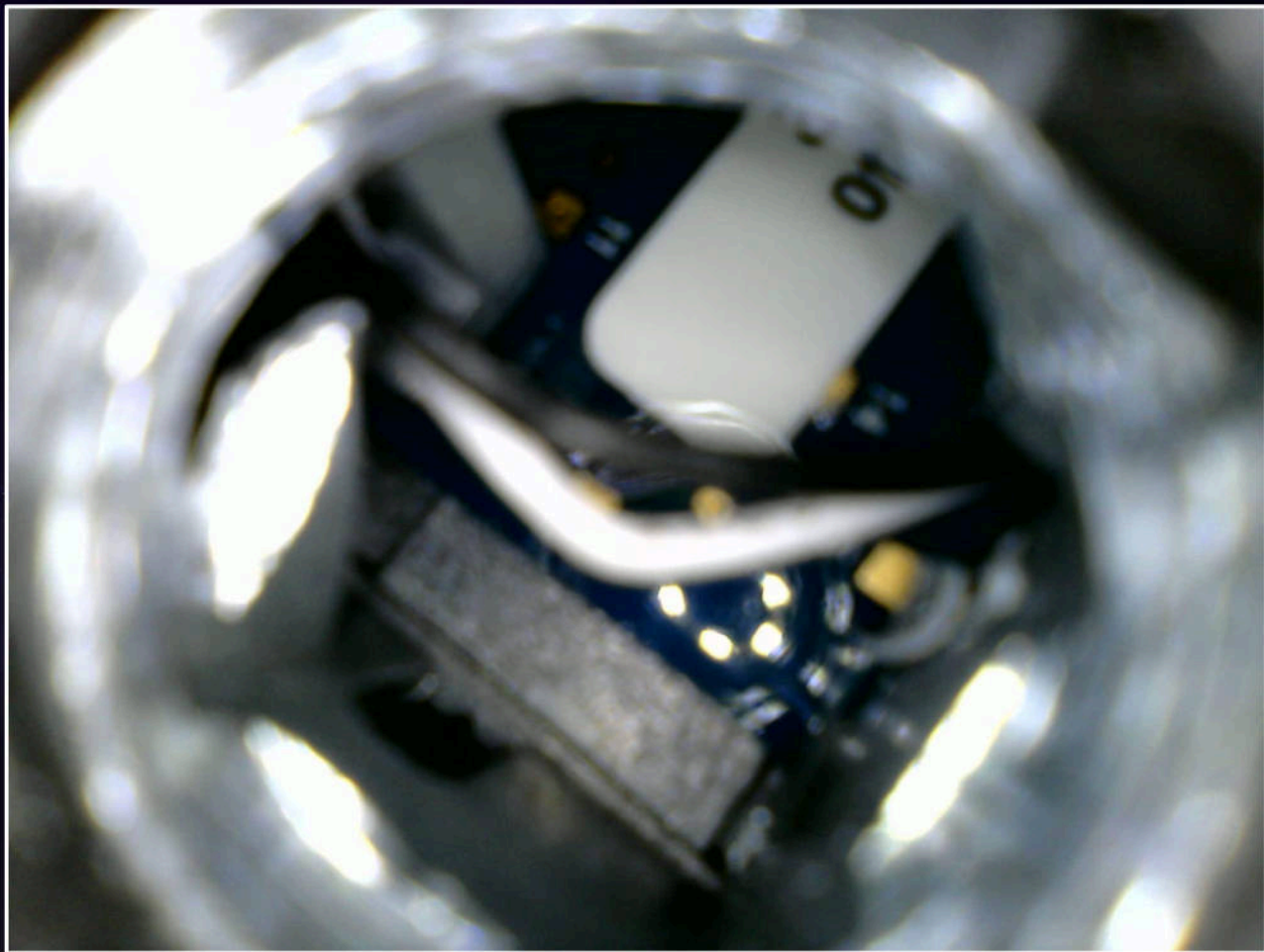




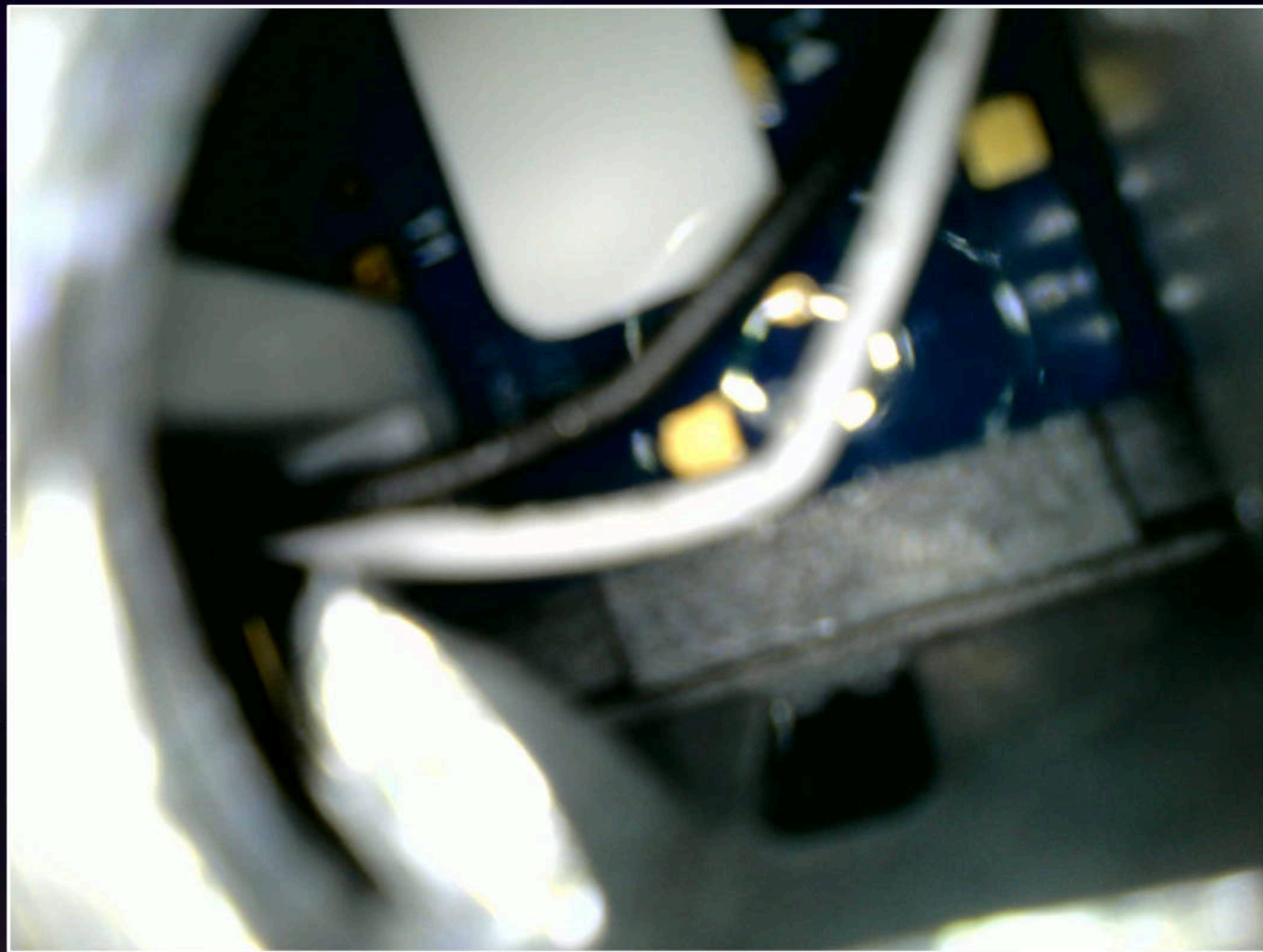




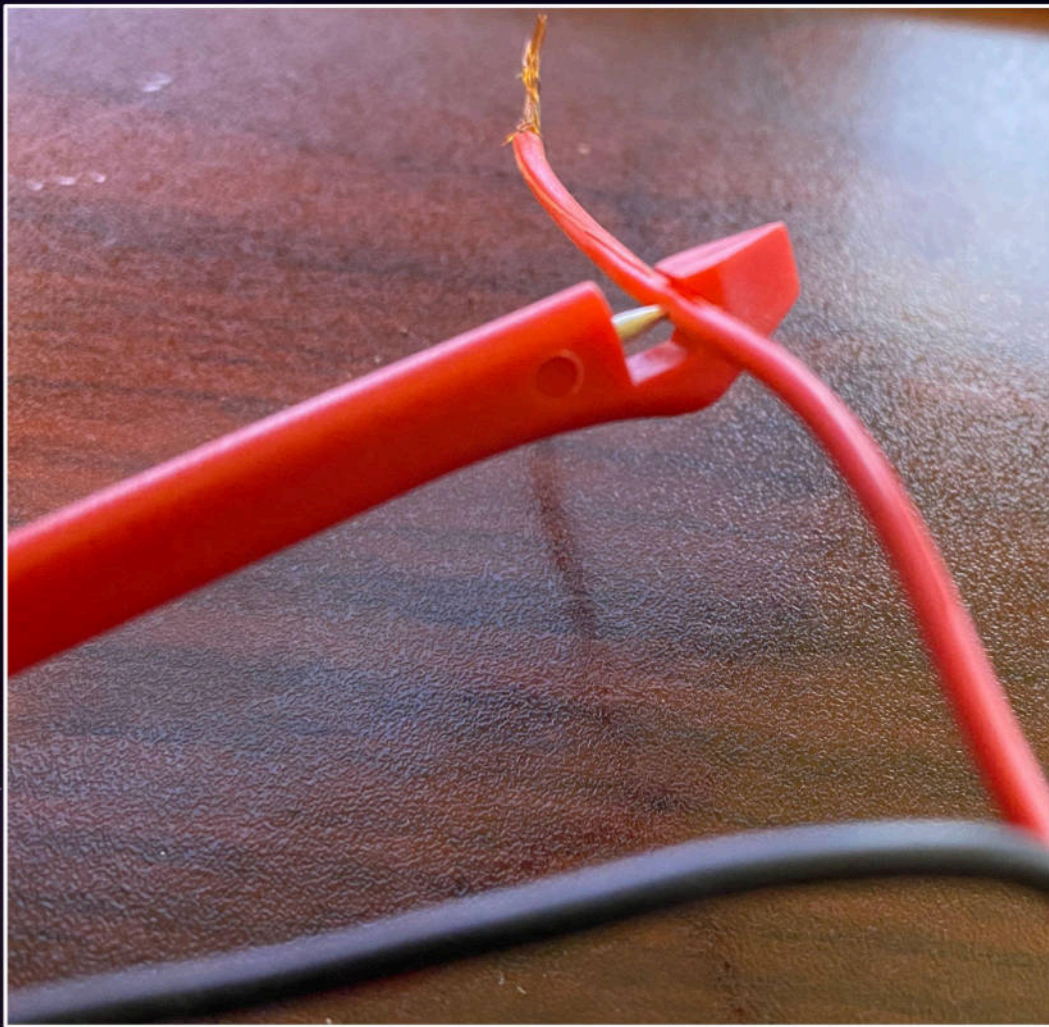


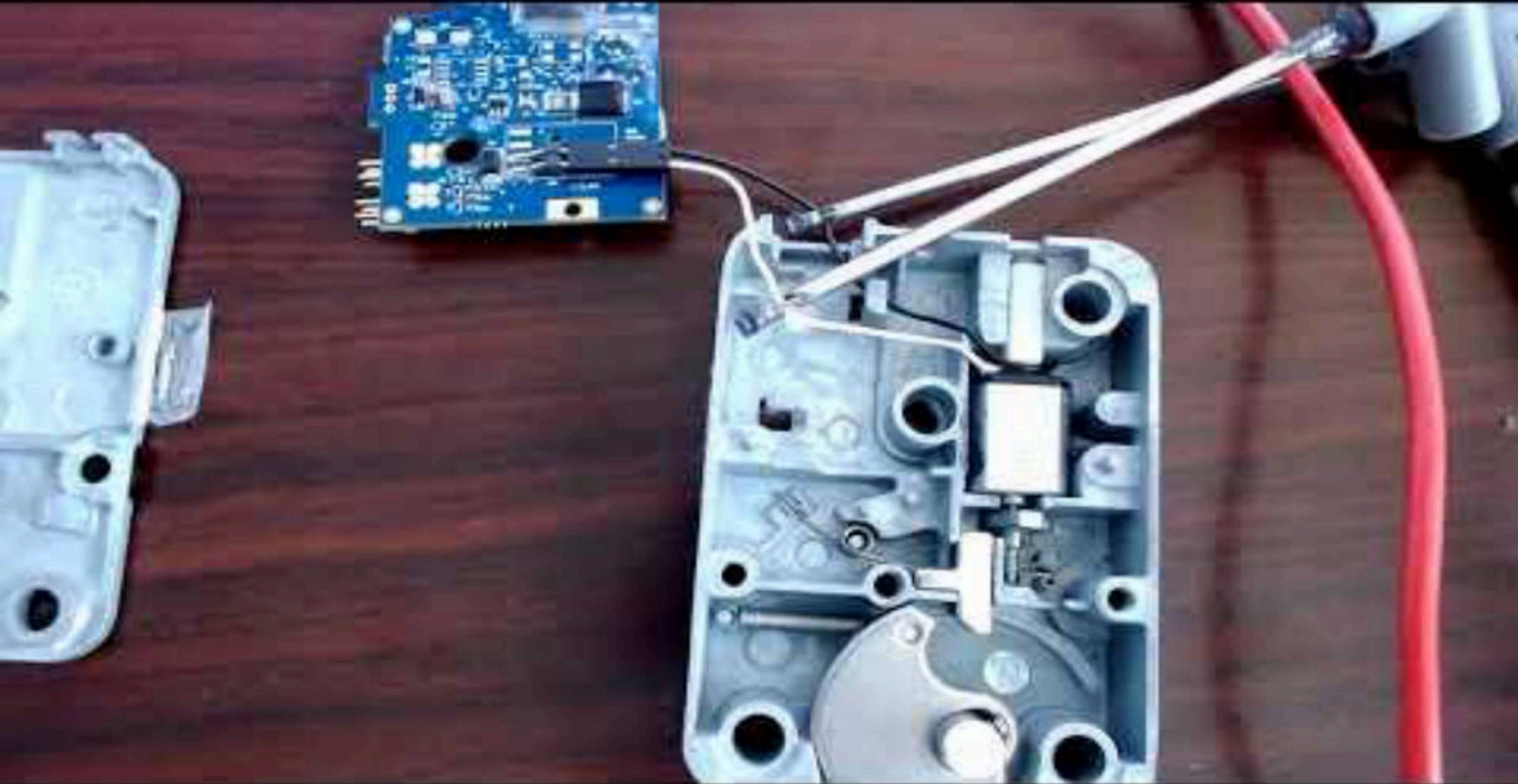


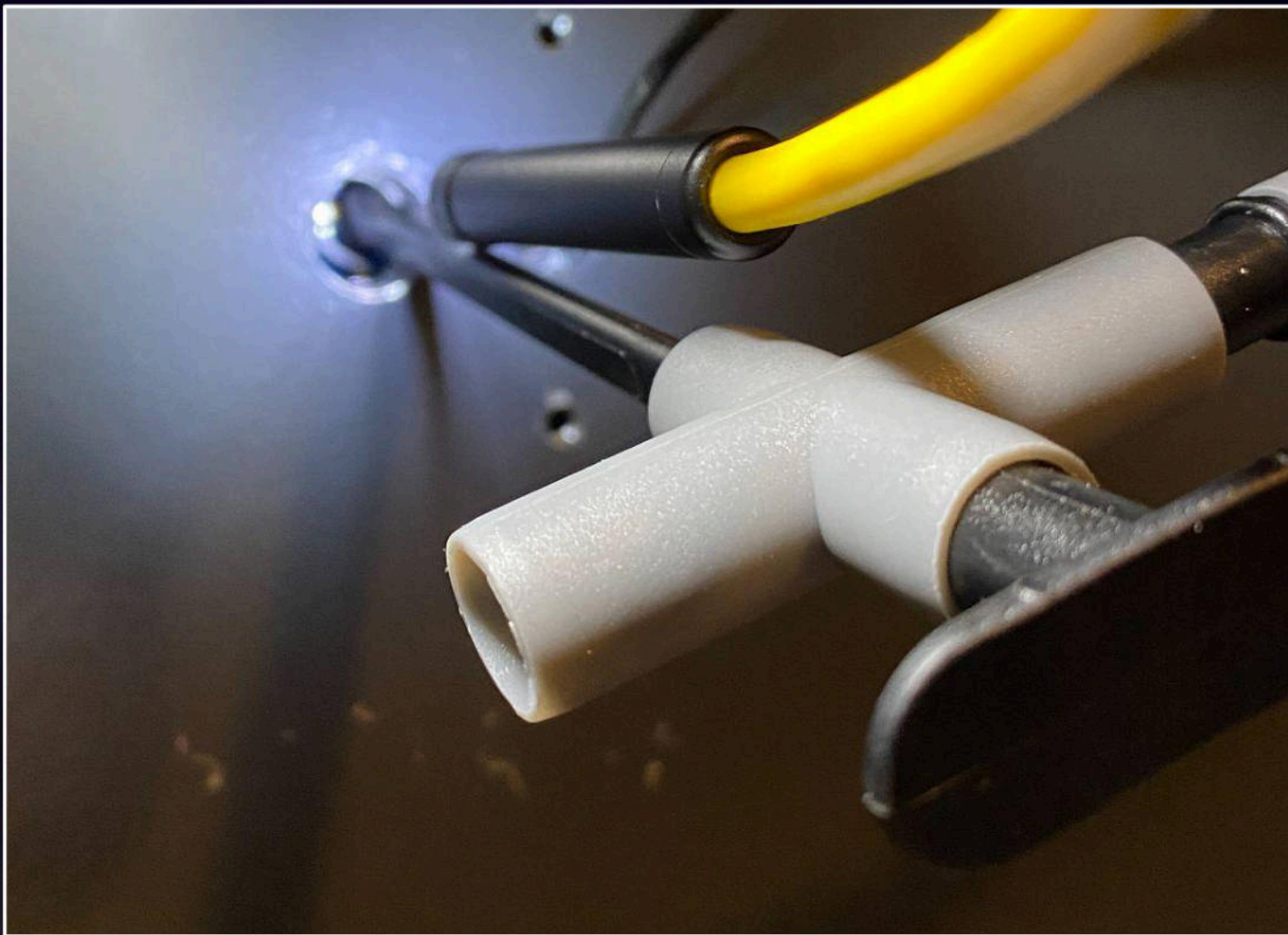


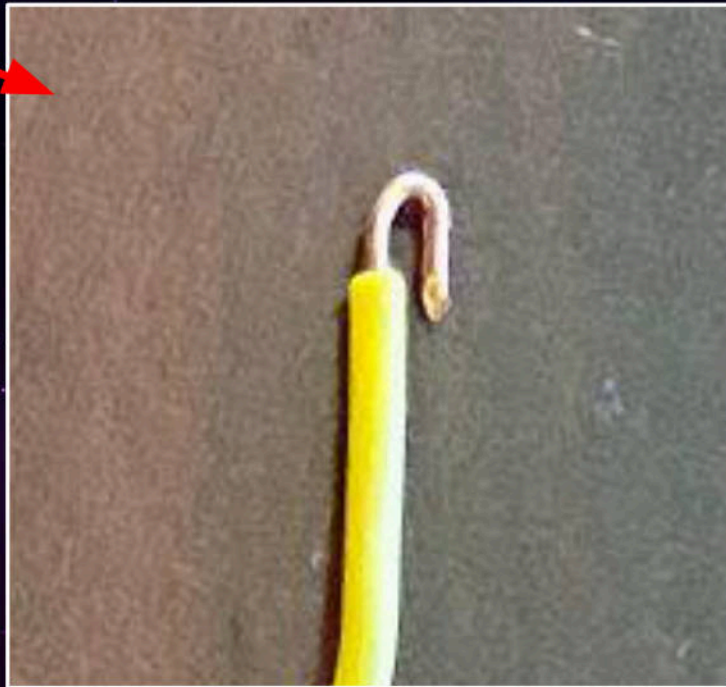
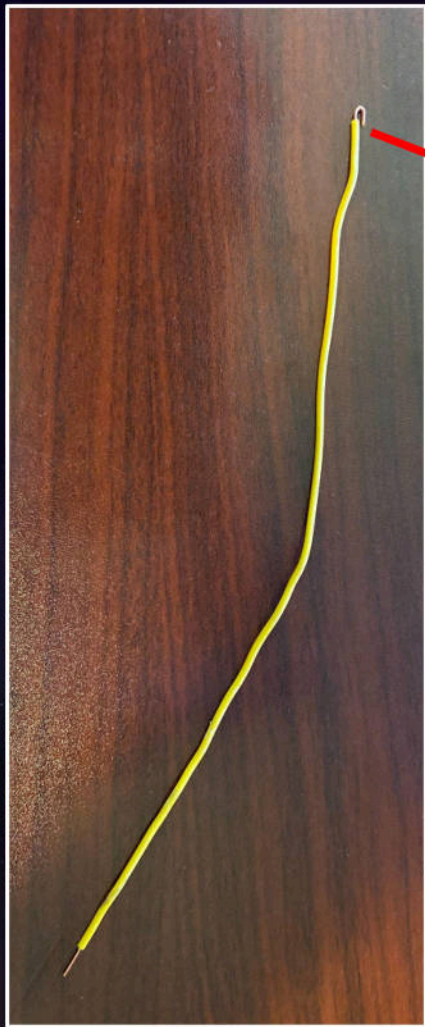


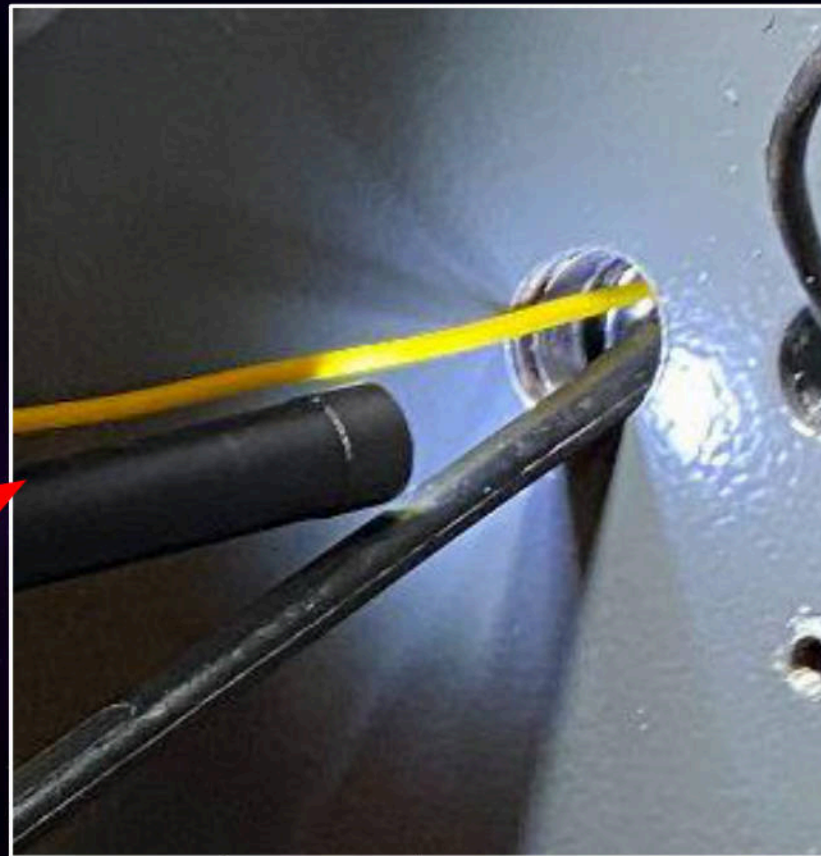
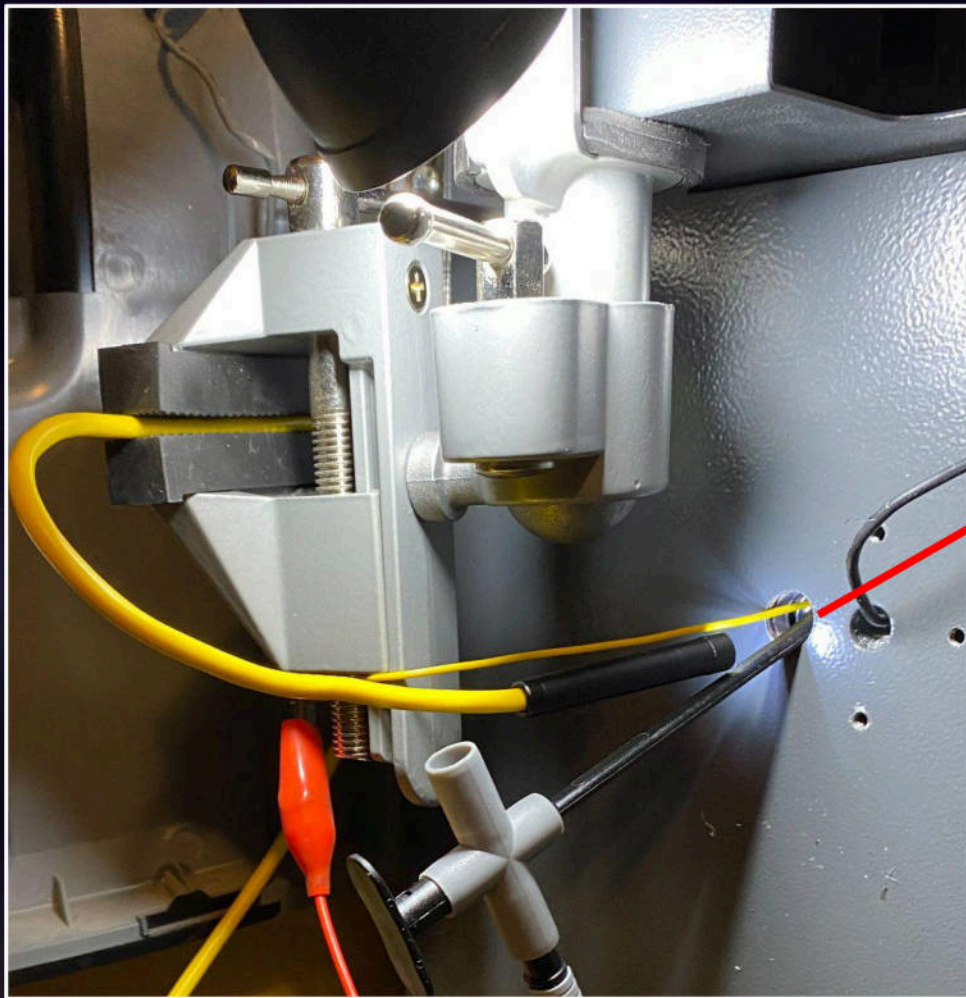


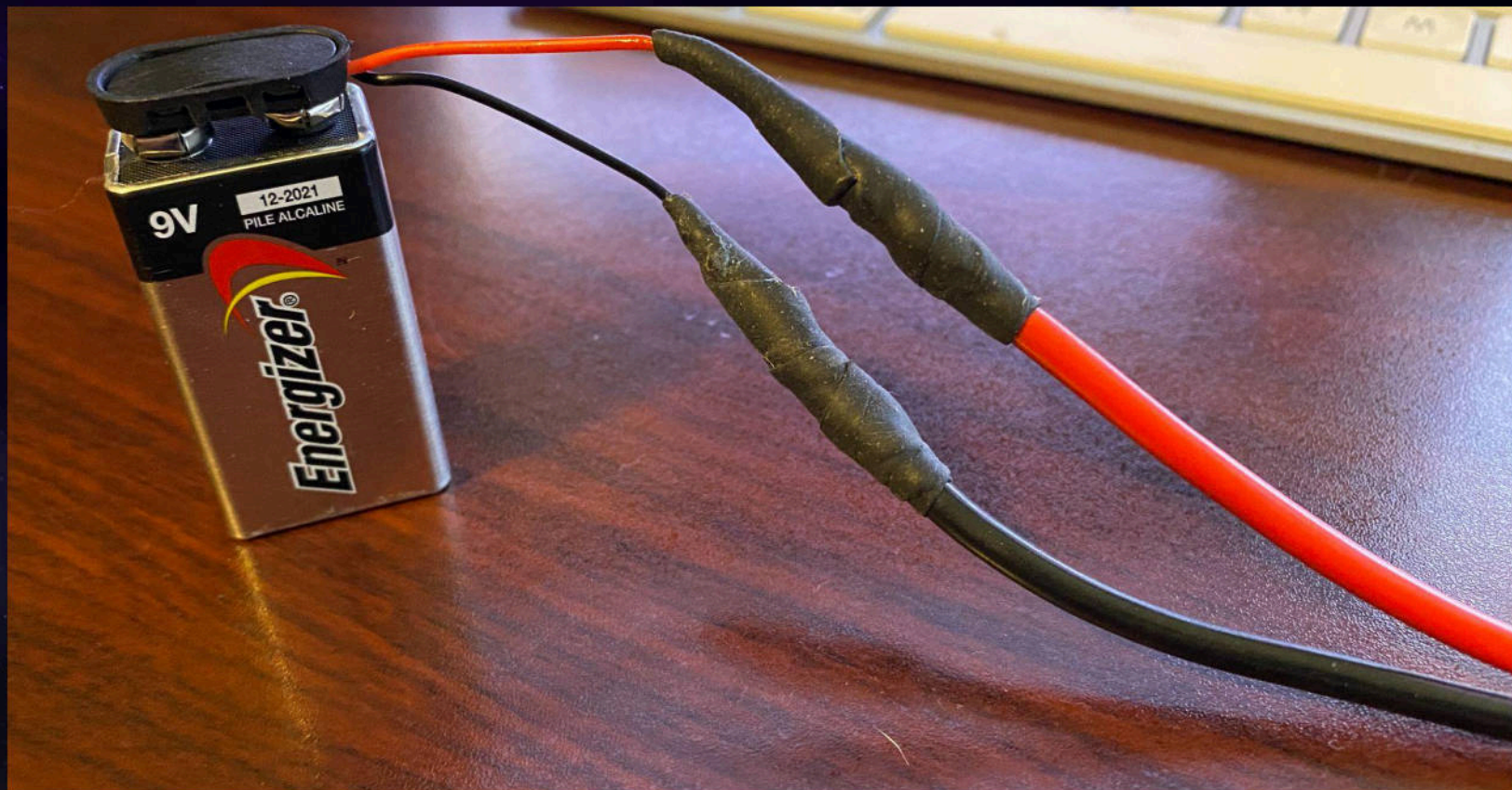










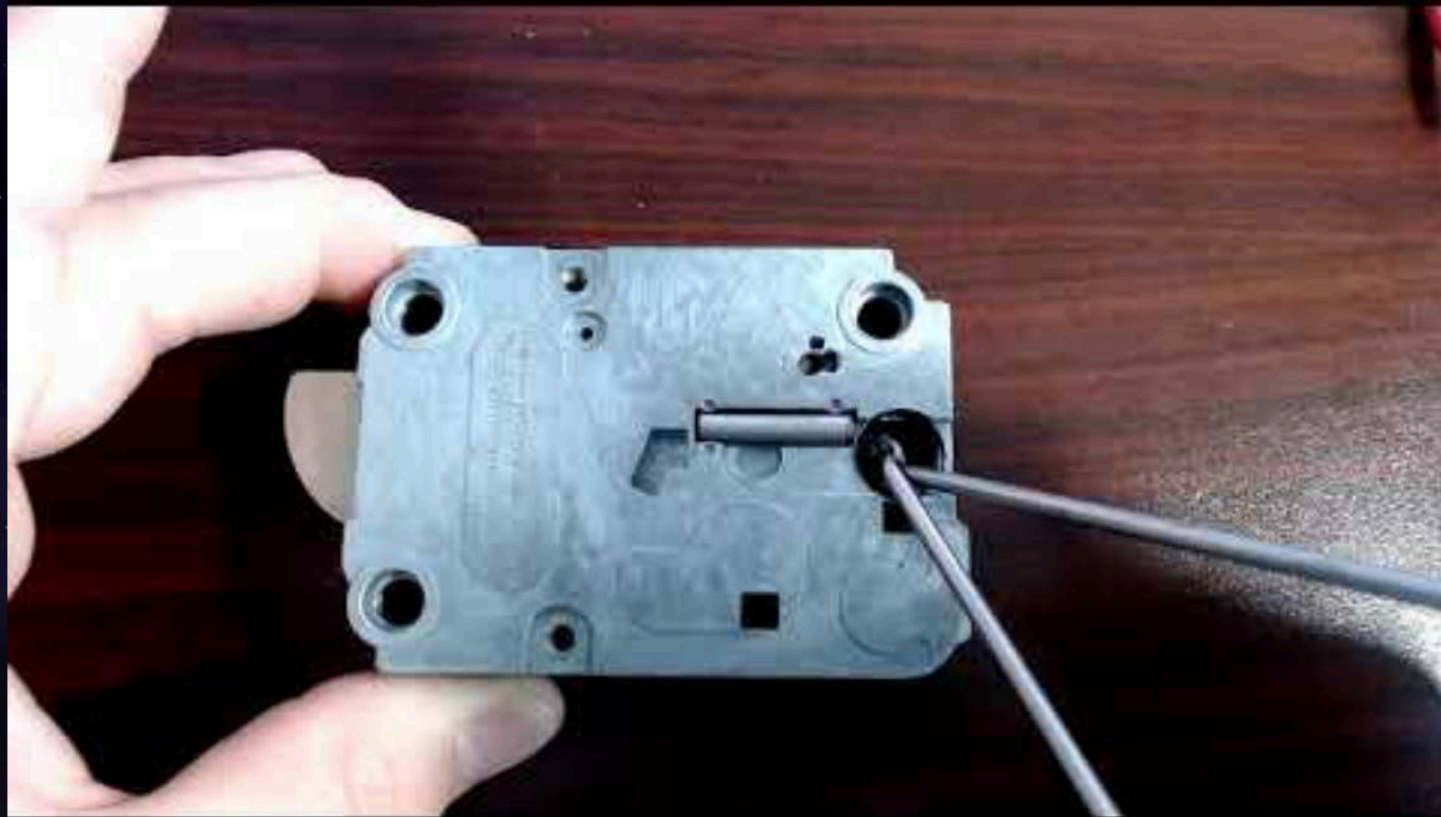


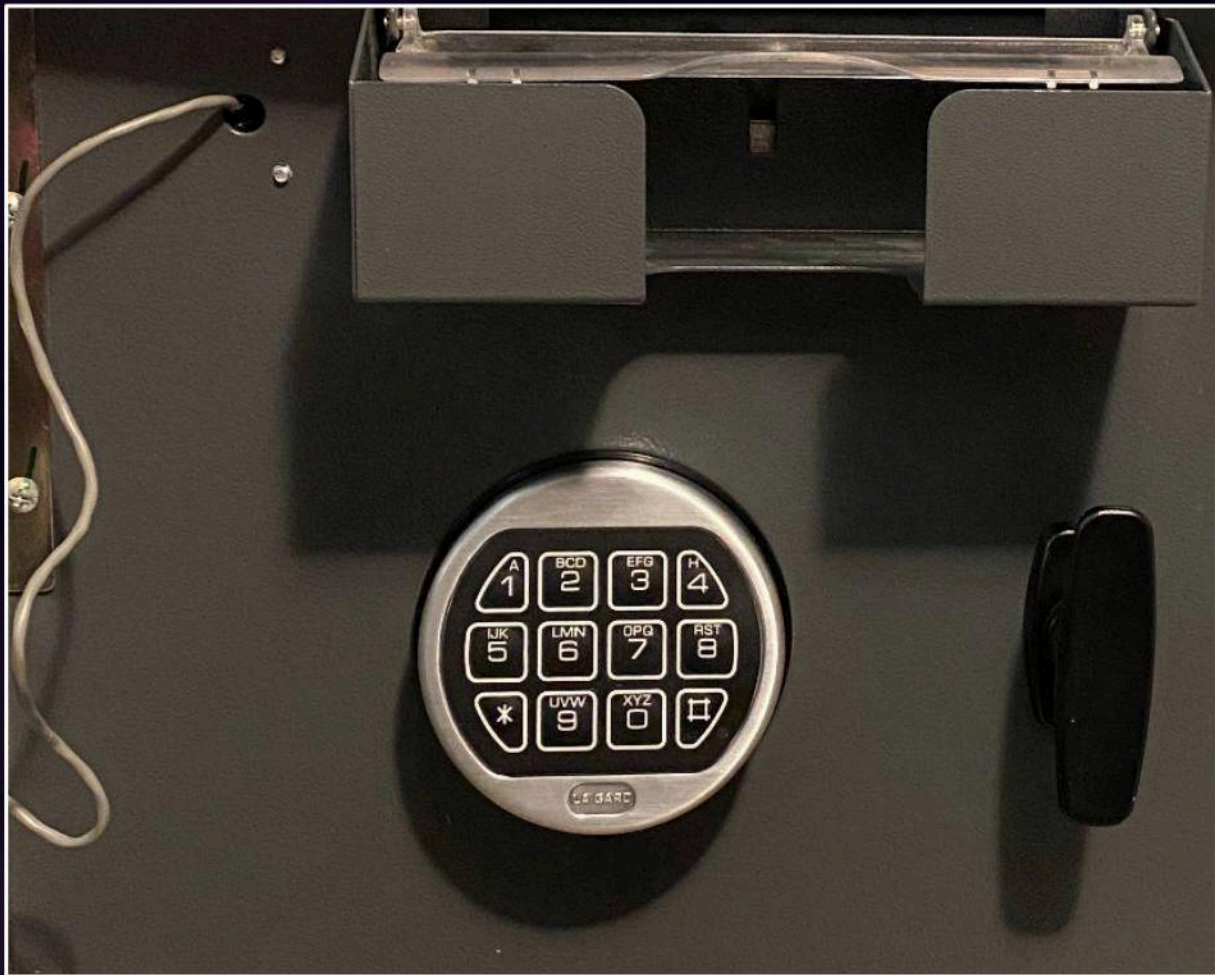


17.7 Volts









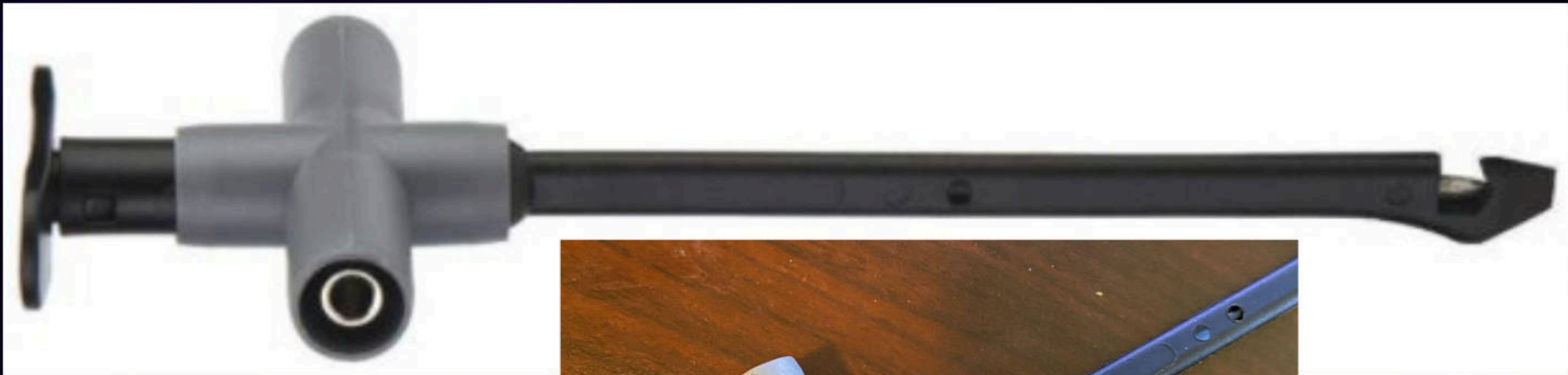


X001J50RIV

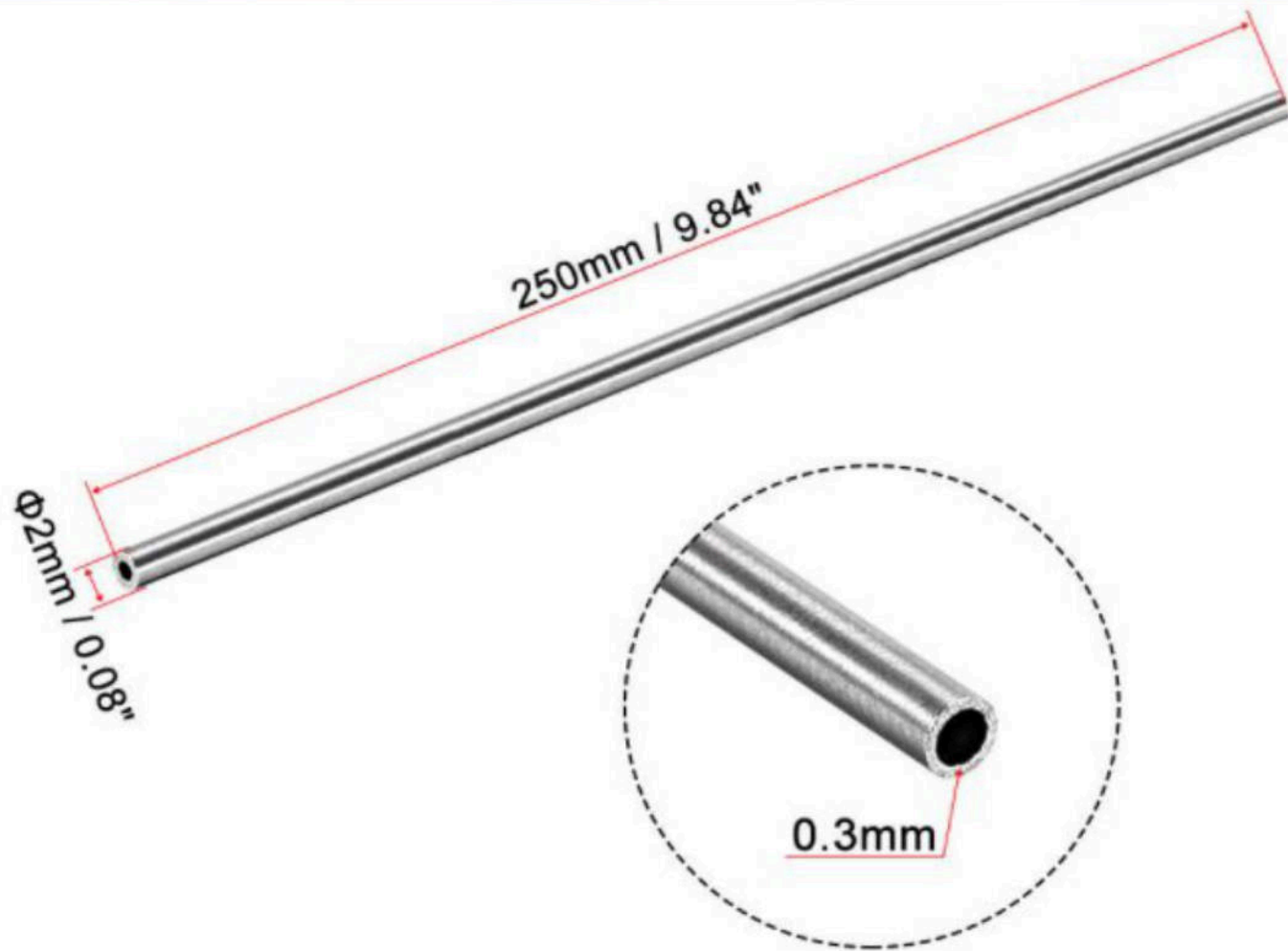
Baomain Plastic Black Loc...ter 1/2" (12.6mm) 100 Pack
New

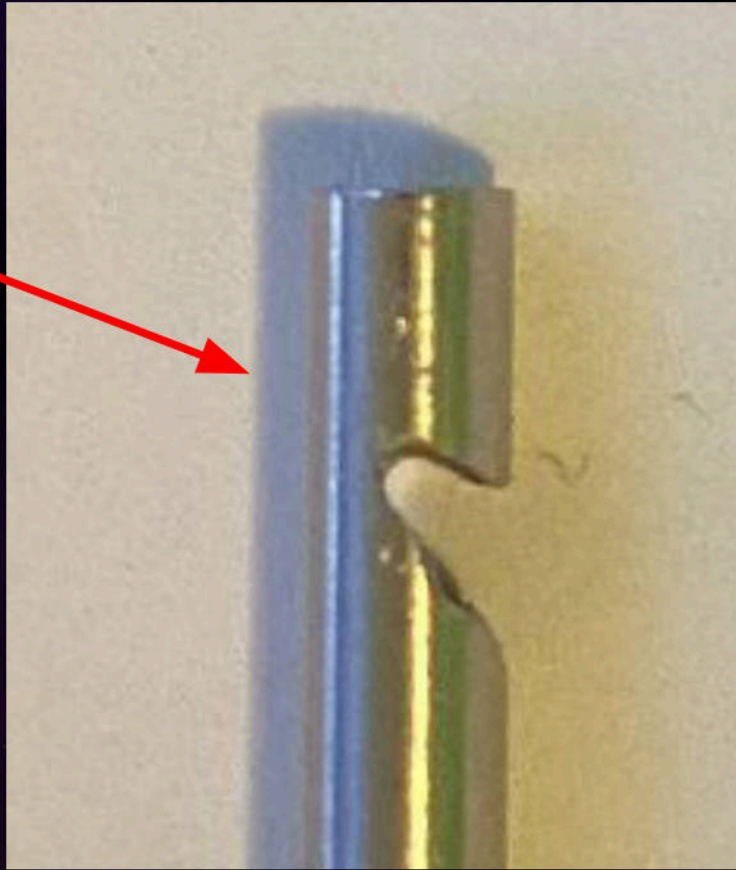
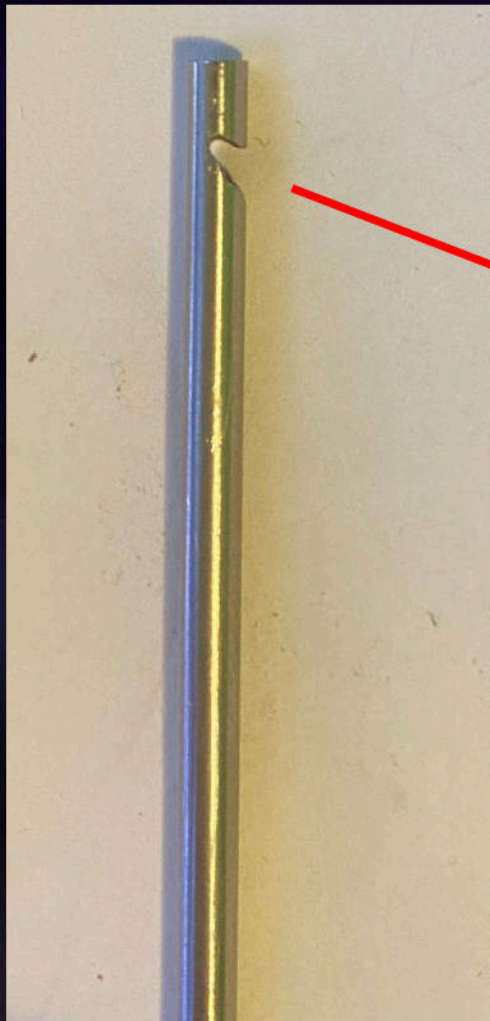






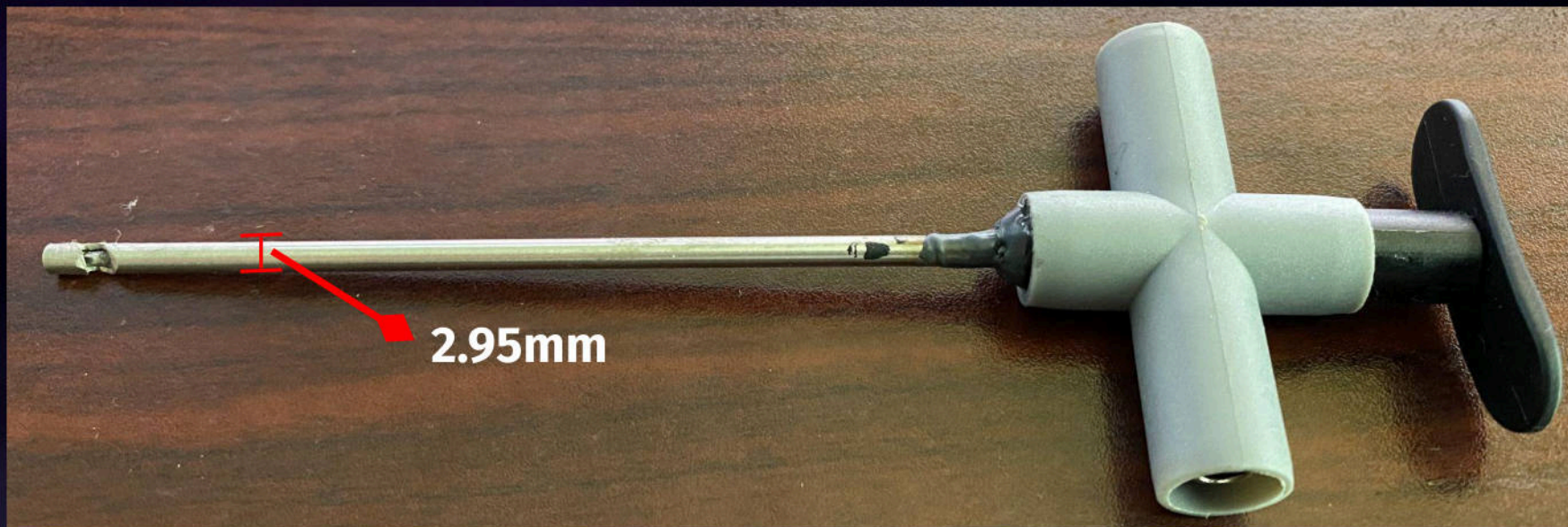














- ☐ Turnstiles and Gates
- ☐ Vacation & Short-term Rental Solutions
- ☐ Workforce Solutions (b-comm, time clocks)
- ☒ I don't know/ other

Please describe the issue

Hello,

I am a security researcher from Minnesota, US
will be giving a talk at DEF CON 29 in Las Vegas
attack that allows bypass of the combination
such a way that no physical evidence of bypo
outside of the vault.

Let me know if you'd like to learn more prior to

Thanks

- Roy Davis

Hello,

I am a security researcher from Minnesota, USA. I am writing to inform you that I
will be giving a talk at DEF CON 29 in Las Vegas which includes a physical
attack that allows bypass of the combination entry for the La Gard Basic, is
such a way that no physical evidence of bypass is readily visible from the
outside of the vault.

Let me know if you'd like to learn more prior to the talk.

Thanks

- Roy Davis



The header of the email interface includes a row of navigation icons: a single left arrow, a double left arrow, a single right arrow, and an ellipsis (three dots).

Roy Davis

Mon 4/26/2021 3:11 PM

To: security@hyosungamericas.com

Hello Hyosung Security Team,

I have identified an attack chain that allows access to the cash inside the Hyosung NH-2700 without having the access key, access code, or safe code, in such a way as to cause no visible damage or evidence of access. I am planning to submit a security conference talk to describe the process and potential remediations. Please contact me ASAP. I would like to provide the details to Hyosung prior to their release.

Roy Davis

Principal Consultant

Davis Information Security Consulting



RD

Roy Davis

Mon 4/26/2021 3:11 PM

To: security@hyosungamericas.com

Hello Hyosung Security Team,

I have identified an attack chain that access to the cash inside the Hyosung 2700 without having the access key, code, or safe code, in such a way as no visible damage or evidence of access. I am planning to submit a security conference talk to describe the process and potential remediations. Please contact me ASAP. I would like to provide the details to Hyosung prior to their release.

Roy Davis

Principal Consultant

Davis Information Security Consulting

MO

Microsoft Outlook

Tue 4/27/2021 3:19 PM

To: Microsoft Outlook



Notification of Security R...

36 KB



Delivery has failed to these recipients or groups:

security@hyosungamericas.com
(security@hyosungamericas.com)

Your message wasn't delivered. Despite repeated attempts to deliver your message, the recipient's email system refused to accept a connection from your email system.





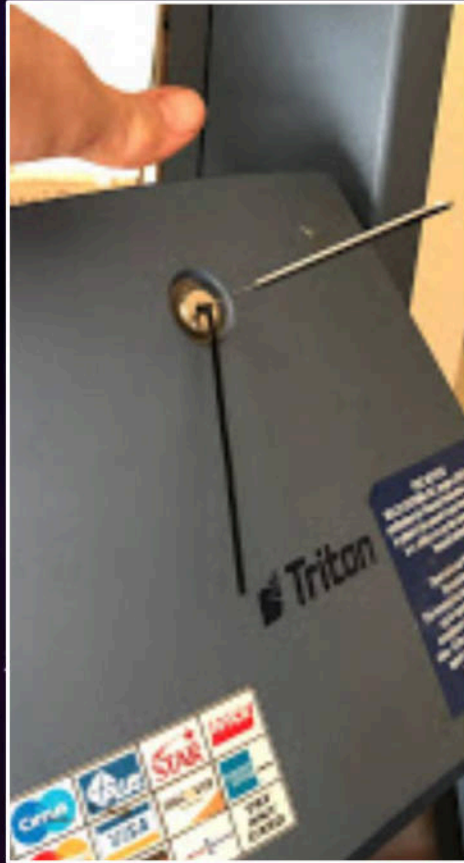


TODO: Follow up Research

- ATM Wifi
- Vault lock MiTM
- ATM Software modifications
- USB and SD Card fun
- Internal Serial comms MiTM and replay
- EPP deconstruction analysis (Warning ??)

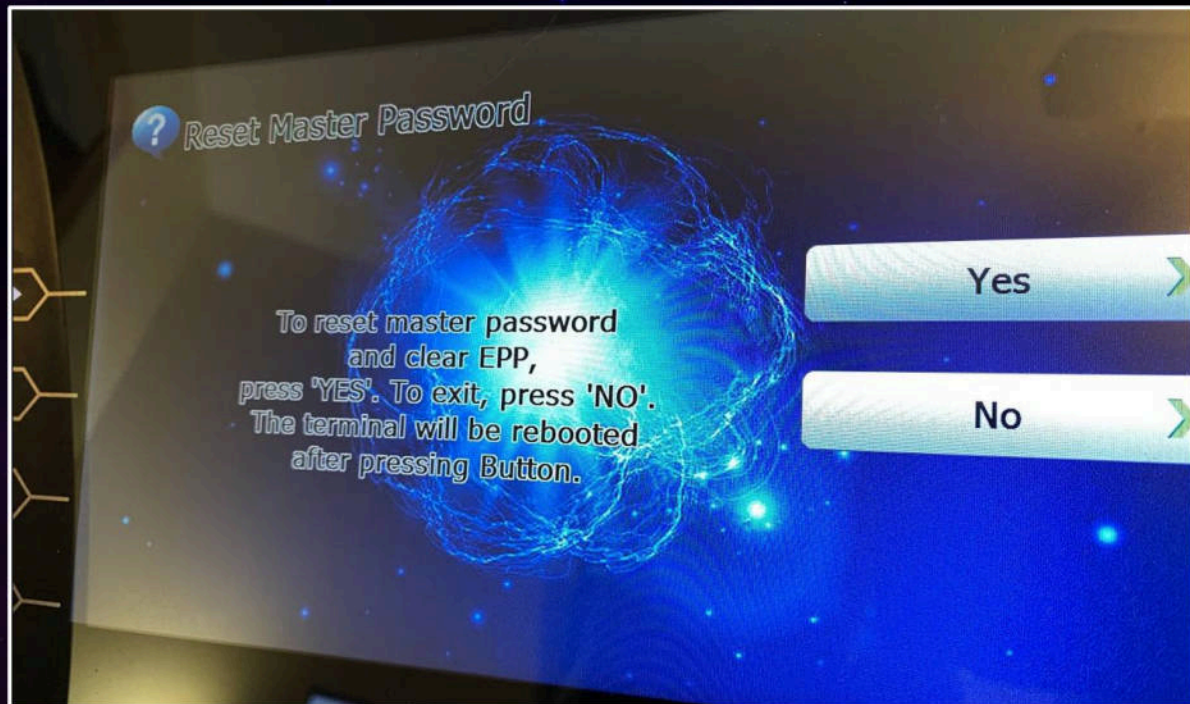


No Key?



No Key?

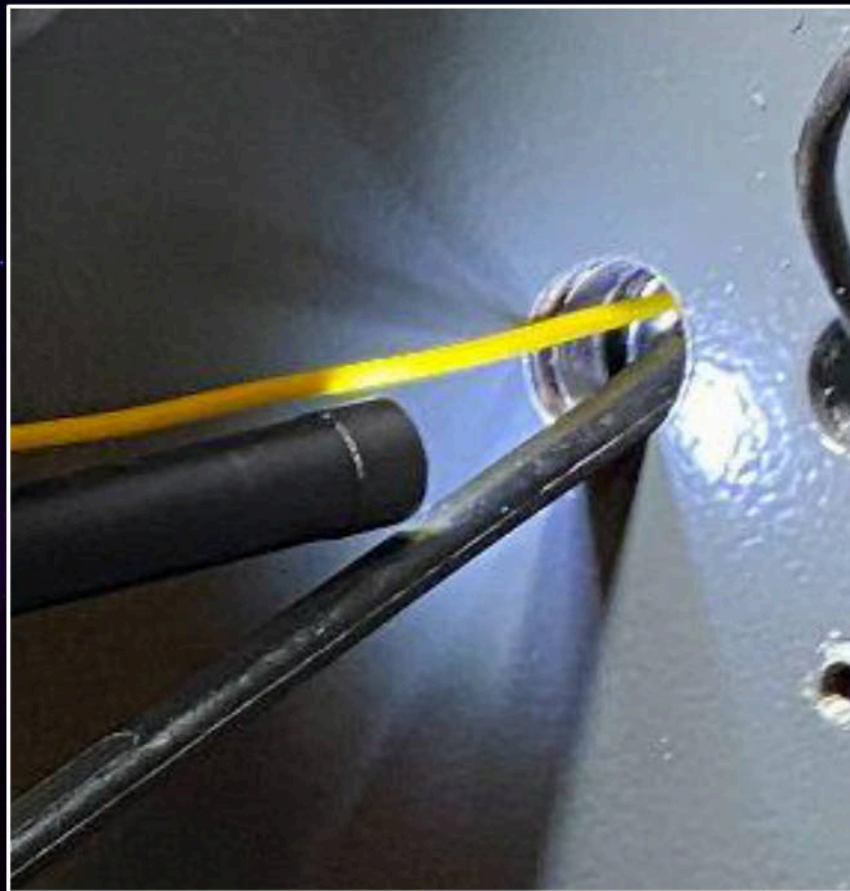
No PIN?



No Key?

No PIN?

No Combo?



No Key?

No PIN?

No Combo?

No Problem!



Thank you!

Roy Davis



roydavis@roydavis.org



[@hack_all_things](https://twitter.com/hack_all_things)



[/in/roy-davis](https://www.linkedin.com/company/roy-davis/)

